Tongji Calculus, 7th Edition · Chapter 5

Definite Integrals

Detailed Solutions

100 classic-method adaptations and high-quality synthesis problems

All mathematics compiled by XeTeX / LaTeX · 4:3 study edition

How to Use This Workbook

The material progresses from basic to challenging. IDs Q001–Q100 match across both languages and both document types. Every formula comes from explicit LaTeX source and is compiled by the XeTeX mathematics engine.

Open-text method adaptations

20 / 100

Classic-method variants

50 / 100

Original synthesis / diagnosis

30 / 100

Source-lineage policy

Problems are redesigned around classic methods found in open textbooks, public courses, and standard calculus teaching. Copyrighted books are used only for scope and method alignment; their wording and solutions are not copied. See SOURCES.md.

1. On the first attempt, use only the exercise workbook and show complete reasoning.
2. During correction, record the cause of each error instead of copying the final answer.
3. Redo errors after 48 hours and interleave topics one week later.

Basic | Single choice | Foundation | Open-text method adaptation

Q001 · Recognizing a Riemann Sum on Equal Subintervals

Problem

Partition $[0, 2]$ into n equal pieces and use right endpoints. Which expression is a Riemann sum for $\int_0^2 (1 + x^2)\,dx$?

A. $\sum_{k=1}^{n} \left[1 + \left(\frac{k}{n}\right)^2\right] \frac{1}{n}$

B. $\sum_{k=1}^{n} \left[1 + \left(\frac{2k}{n}\right)^2\right] \frac{2}{n}$

C. $\sum_{k=1}^{n} \left[1 + \left(\frac{2k}{n}\right)^2\right]$

D. $\sum_{k=0}^{n} \left[1 + \left(\frac{2k}{n}\right)^2\right] \frac{2}{n}$

Answer

B

Knowledge points

- definition by Riemann sums
- subinterval widths and sample points
- limits of sums and definite integrals

Reading and method choice

The interval gives $\Delta x = \frac{2}{n}$ and right endpoint $x_k = \frac{2k}{n}$; the function value must be multiplied by the width.

Detailed derivation

1. Equal subdivision gives $\Delta x = \frac{2-0}{n} = \frac{2}{n}$.
2. The kth right endpoint is $x_k = 0 + k\Delta x = \frac{2k}{n}$.
3. The sampled value is $1 + x_k^2 = 1 + \left(\frac{2k}{n}\right)^2$.
4. Hence the sum is $\sum_{k=1}^{n} \left[1 + \left(\frac{2k}{n}\right)^2\right] \frac{2}{n}$, which is B.

Common pitfalls

- omitting the width factor Δx
- failing to check both sample points and the integration interval

Verification

Checking the constant part 1 gives $n \cdot \frac{2}{n} = 2$; only B retains the correct width.

Method takeaway

A Riemann sum is identified only after matching the function value, sample points, subinterval width, and interval.

Extension

Left endpoints or midpoints may be used instead; continuity gives the same limiting integral.

Source lineage: Open-text method adaptation · riemann_sums_and_definition · reference IDs: mit-18.01sc-definite-integral, openstax-calculus-v1-5.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic Single choice Foundation Open-text method adaptation

Q002 · Distinguishing an Integral from Geometric Area

Problem

If a continuous function satisfies $f(x) \leq 0$ on $[a, b]$, which statement is correct?

A. $\int_a^b f(x)\,dx \geq 0$

B. The geometric area equals $\int_a^b f(x)\,dx$.

C. The geometric area equals $-\int_a^b f(x)\,dx$.

D. The definite integral must equal 0.

Answer

C

Knowledge points

- linearity of definite integrals
- interval additivity and reversal
- order bounds and symmetry

Reading and method choice

A definite integral records signed area. A graph below the horizontal axis gives a nonpositive integral, while geometric area is its negative.

Detailed derivation

1. From $f(x) \leq 0$ and order preservation, $\int_a^b f(x)\,dx \leq 0$.
2. Geometric area between the graph and the horizontal axis is nonnegative.
3. Under the stated sign condition, that area equals $-\int_a^b f(x)\,dx$.
4. Thus only C gives both the correct sign and the correct area relation.

Common pitfalls

- identifying a definite integral with unsigned geometric area
- forgetting the sign change when limits are reversed

Verification

For $f(x) = -1$, the integral is $-(b-a)$ while the geometric area is $b-a$, confirming C.

Method takeaway

Check sign, interval orientation, and symmetry before carrying out piecewise computation.

Extension

Splitting an integrand into odd and even parts can further simplify integrals over symmetric intervals.

Source lineage: Open-text method adaptation · definite_integral_properties · reference IDs: mit-18.01sc-definite-integral, openstax-calculus-v1-5.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic | Single choice | Foundation | Open-text method adaptation

Q003 · Reversing the Limits of Integration

Problem

Given $\int_1^4 f(x)\,dx = 7$, what is $\int_4^1 f(x)\,dx$?

A. -7

B. 0

C. 7

D. 14

Answer

A

Knowledge points

- linearity of definite integrals
- interval additivity and reversal
- order bounds and symmetry

Reading and method choice

Reversing the limits changes only the orientation, not the integrand or geometric interval, and therefore multiplies the integral by -1.

Detailed derivation

1. The reversal rule is $\int_b^a f(x)\,dx = -\int_a^b f(x)\,dx$.
2. Set $a = 1$ and $b = 4$.
3. Using the given value gives $\int_4^1 f(x)\,dx = -7$.
4. Therefore choice A is the unique correct option.

Common pitfalls

- identifying a definite integral with unsigned geometric area
- forgetting the sign change when limits are reversed

Verification

The oppositely oriented integrals add to $7 + (-7) = 0$, consistent with $\int_1^1 f(x)\,dx = 0$.

Method takeaway

Check sign, interval orientation, and symmetry before carrying out piecewise computation.

Extension

Splitting an integrand into odd and even parts can further simplify integrals over symmetric intervals.

Source lineage: Open-text method adaptation · definite_integral_properties · reference IDs: mit-18.01sc-definite-integral, openstax-calculus-v1-5.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic | Single choice | Foundation | Open-text method adaptation

Q004 · Bounding a Definite Integral

Problem

If f is integrable on $[1,4]$ and $2 \le f(x) \le 5$ for every $x \in [1,4]$, which statement must hold?

A. $2 \le \int_1^4 f(x)\,dx \le 5$

B. $3 \le \int_1^4 f(x)\,dx \le 12$

C. $5 \le \int_1^4 f(x)\,dx \le 20$

D. $6 \le \int_1^4 f(x)\,dx \le 15$

Answer

D

Knowledge points

- linearity of definite integrals
- interval additivity and reversal
- order bounds and symmetry

Reading and method choice

Integrating the pointwise bounds over an interval of length 3 gives two-sided bounds for the integral.

Detailed derivation

1. Order preservation gives $\int_1^4 2\,dx \le \int_1^4 f(x)\,dx \le \int_1^4 5\,dx$.
2. The lower bound is $2(4-1) = 6$.
3. The upper bound is $5(4-1) = 15$.
4. Therefore $6 \le \int_1^4 f(x)\,dx \le 15$, which is D.

Common pitfalls

- identifying a definite integral with unsigned geometric area
- forgetting the sign change when limits are reversed

Verification

The constant functions $f(x) = 2$ and $f(x) = 5$ attain the two endpoints, so the bounds cannot be uniformly tightened.

Method takeaway

Check sign, interval orientation, and symmetry before carrying out piecewise computation.

Extension

Splitting an integrand into odd and even parts can further simplify integrals over symmetric intervals.

Source lineage: Open-text method adaptation · definite_integral_properties · reference IDs: mit-18.01sc-definite-integral, openstax-calculus-v1-5.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic Multiple choice Foundation Open-text method adaptation

Q005 · Selecting Valid Integral Identities

Problem

Assume all displayed integrals exist. Which identities hold for every integrand?

A. $\int_a^b \lambda f(x)\,dx = \lambda \int_a^b f(x)\,dx$

B. $\int_a^b f(x)g(x)\,dx = \left(\int_a^b f(x)\,dx\right)\left(\int_a^b g(x)\,dx\right)$

C. $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$

D. $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$

Answer

A, C, D

Knowledge points

- linearity of definite integrals
- interval additivity and reversal
- order bounds and symmetry

Reading and method choice

Check linearity, reversal, and interval additivity one by one. The integral of a product cannot generally be split into a product of integrals.

Detailed derivation

1. A is scalar linearity and is valid.
2. B is generally false; on $[0, 1]$, taking $f(x) = g(x) = x$ gives $\frac{1}{3}$ on the left and $\frac{1}{4}$ on the right.
3. C is the reversal formula and is valid.
4. D is interval additivity and is valid for every c.

Common pitfalls

- identifying a definite integral with unsigned geometric area
- forgetting the sign change when limits are reversed

Verification

In the counterexample, $\int_0^1 x^2\,dx = \frac{1}{3}$ whereas $\left(\int_0^1 x\,dx\right)^2 = \frac{1}{4}$, excluding B.

Method takeaway

Check sign, interval orientation, and symmetry before carrying out piecewise computation.

Extension

Splitting an integrand into odd and even parts can further simplify integrals over symmetric intervals.

Source lineage: Open-text method adaptation · definite_integral_properties · reference IDs: mit-18.01sc-definite-integral, openstax-calculus-v1-5.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic | Multiple choice | Foundation | Open-text method adaptation

Q006 · Recognizing Integrability and Order Consequences

Problem

On a closed interval $[a, b]$, which statements are correct?

A. Every continuous function is Riemann integrable.

B. Every monotone function is Riemann integrable.

C. Equal integrals of two continuous functions imply pointwise equality.

D. If $f(x) \geq 0$, then $\int_a^b f(x)\,dx \geq 0$.

Answer

A, B, D

Knowledge points

- linearity of definite integrals
- interval additivity and reversal
- order bounds and symmetry

Reading and method choice

This problem separates standard sufficient conditions, order preservation, and an invalid converse.

Detailed derivation

1. A continuous function on a closed interval is Riemann integrable, so A is correct.
2. A monotone function on a closed interval is bounded with controlled discontinuities and is Riemann integrable, so B is correct.
3. Equal integrals compare only total accumulation and do not imply pointwise equality, so C is false.
4. If $f(x) \geq 0$, every Riemann sum is nonnegative, and its limit is nonnegative; hence D is correct.

Common pitfalls

- identifying a definite integral with unsigned geometric area
- forgetting the sign change when limits are reversed

Verification

For C, take $f(x) = x$ and $g(x) = 1 - x$ on $[0, 1]$. Both integrals equal $\frac{1}{2}$, but the functions are not identical.

Method takeaway

Check sign, interval orientation, and symmetry before carrying out piecewise computation.

Extension

Splitting an integrand into odd and even parts can further simplify integrals over symmetric intervals.

Source lineage: Open-text method adaptation · definite_integral_properties · reference IDs: mit-18.01sc-definite-integral, openstax-calculus-v1-5.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic | True/false with justification | Foundation | Open-text method adaptation

Q007 · The Absolute-Value Integral Inequality

Problem

Determine whether the statement is true and justify: if f is integrable on $[a, b]$, then $\left|\int_a^b f(x)\,dx\right| \le \int_a^b |f(x)|\,dx$.

Answer

True.

Knowledge points

- linearity of definite integrals
- interval additivity and reversal
- order bounds and symmetry

Reading and method choice

Start with the pointwise inequality $-|f(x)| \le f(x) \le |f(x)|$ and integrate to trap the original integral.

Detailed derivation

1. For every $x \in [a, b]$, $-|f(x)| \le f(x) \le |f(x)|$.
2. Order preservation gives $-\int_a^b |f(x)|\,dx \le \int_a^b f(x)\,dx \le \int_a^b |f(x)|\,dx$.
3. Being trapped between the negative and positive of the same nonnegative number is equivalent to $\left|\int_a^b f(x)\,dx\right| \le \int_a^b |f(x)|\,dx$.
4. Hence the statement is true; equality depends on whether positive and negative contributions cancel.

Common pitfalls

- identifying a definite integral with unsigned geometric area
- forgetting the sign change when limits are reversed

Verification

Equality holds when f has constant sign; cancellation between positive and negative parts makes the left side smaller, as expected.

Method takeaway

Check sign, interval orientation, and symmetry before carrying out piecewise computation.

Extension

Splitting an integrand into odd and even parts can further simplify integrals over symmetric intervals.

Source lineage: Open-text method adaptation · definite_integral_properties · reference IDs: mit-18.01sc-definite-integral, openstax-calculus-v1-5.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic | True/false with justification | Foundation | Open-text method adaptation

Q008 · Inferring Equality of Functions from Equal Integrals

Problem

Determine whether the statement is true and justify: if continuous functions f, g satisfy $\int_0^1 f(x)\,dx = \int_0^1 g(x)\,dx$, then $f(x) = g(x)$ throughout $[0, 1]$.

Answer

False.

Knowledge points

- linearity of definite integrals
- interval additivity and reversal
- order bounds and symmetry

Reading and method choice

A single integral gives only one aggregate constraint and cannot determine a whole function; two distinct functions with the same average provide a counterexample.

Detailed derivation

1. Take $f(x) = x$ and $g(x) = 1 - x$; both are continuous on $[0, 1]$.
2. We have $\int_0^1 x\,dx = \frac{1}{2}$.
3. Also $\int_0^1 (1 - x)\,dx = \frac{1}{2}$.
4. Yet at $x = 0$, $f(0) = 0 \neq 1 = g(0)$, so the statement is false.

Common pitfalls

- identifying a definite integral with unsigned geometric area
- forgetting the sign change when limits are reversed

Verification

The counterexample satisfies continuity and equality of integrals but not equality of functions, which disproves the universal statement.

Method takeaway

Check sign, interval orientation, and symmetry before carrying out piecewise computation.

Extension

Splitting an integrand into odd and even parts can further simplify integrals over symmetric intervals.

Source lineage: Open-text method adaptation · definite_integral_properties · reference IDs: mit-18.01sc-definite-integral, openstax-calculus-v1-5.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic | True/false with justification | Foundation | Open-text method adaptation

Q009 · Symmetric Integral of an Odd Function

Problem

Determine whether the statement is true and justify: if f is an integrable odd function on $[-a, a]$, then $\int_{-a}^{a} f(x)\,dx = 0$.

Answer

True.

Knowledge points

- linearity of definite integrals
- interval additivity and reversal
- order bounds and symmetry

Reading and method choice

Split the integral at 0 and substitute $x = -u$ in the negative-half integral.

Detailed derivation

1. Interval additivity gives $\int_{-a}^{a} f(x)\,dx = \int_{-a}^{0} f(x)\,dx + \int_{0}^{a} f(x)\,dx$.
2. In the first term let $x = -u$; then $dx = -du$ and $\int_{-a}^{0} f(x)\,dx = \int_{0}^{a} f(-u)\,du$.
3. Oddness gives $f(-u) = -f(u)$, so the first term is $-\int_{0}^{a} f(u)\,du$.
4. The two terms cancel, giving total integral 0.

Common pitfalls

- identifying a definite integral with unsigned geometric area
- forgetting the sign change when limits are reversed

Verification

For $f(x) = x^3$, the two symmetric parts have opposite signs and equal magnitude, giving 0.

Method takeaway

Check sign, interval orientation, and symmetry before carrying out piecewise computation.

Extension

Splitting an integrand into odd and even parts can further simplify integrals over symmetric intervals.

Source lineage: Open-text method adaptation · definite_integral_properties · reference IDs: mit-18.01sc-definite-integral, openstax-calculus-v1-5.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic Fill in the blank Foundation Open-text method adaptation

Q010 · A Definite Integral with Equal Limits

Problem

For every integrable function f, $\int_a^a f(x)\,dx =$ ____.

Answer

0

Knowledge points

- linearity of definite integrals
- interval additivity and reversal
- order bounds and symmetry

Reading and method choice

An interval of zero length has no accumulation; the same result follows immediately from the reversal rule.

Detailed derivation

1. The reversal rule gives $\int_a^a f(x)\,dx = -\int_a^a f(x)\,dx$.
2. Adding the two sides yields $2\int_a^a f(x)\,dx = 0$.
3. Therefore $\int_a^a f(x)\,dx = 0$.
4. Thus the blank is 0, independently of the particular integrable function f.

Common pitfalls

- identifying a definite integral with unsigned geometric area
- forgetting the sign change when limits are reversed

Verification

The total interval length in the Riemann-sum definition is $a - a = 0$, consistent with the answer.

Method takeaway

Check sign, interval orientation, and symmetry before carrying out piecewise computation.

Extension

Splitting an integrand into odd and even parts can further simplify integrals over symmetric intervals.

Source lineage: Open-text method adaptation · definite_integral_properties · reference IDs: mit-18.01sc-definite-integral, openstax-calculus-v1-5.2. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic | Fill in the blank | Foundation | Classic-method variant

Q011 · Average Value of a Constant Function

Problem

The average value of $f(x) = 7$ on $[-2, 3]$ is ____.

Answer

7

Knowledge points

- average value of a function
- mean value theorem for integrals
- range of a continuous function

Reading and method choice

Apply the average-value formula and use that the integral of a constant is the constant times the interval length.

Detailed derivation

1. The interval length is $3 - (-2) = 5$.
2. The average value is $f_{\text{avg}} = \frac{1}{5}\int_{-2}^{3} 7\,dx$.
3. The integral is $7 \cdot 5 = 35$, so $f_{\text{avg}} = \frac{35}{5} = 7$.
4. Thus the blank is 7; the interval length cancels between numerator and denominator.

Common pitfalls

- forgetting to divide by the interval length
- assuming the mean-value point is unique

Verification

The function equals 7 at every point, so its average cannot differ from 7.

Method takeaway

The average value is the integral divided by interval length; the mean value theorem guarantees at least one corresponding point.

Extension

If the function is strictly monotone, the point satisfying the mean-value identity is unique.

Source lineage: Classic-method variant · integral_mean_value_and_average · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic | Fill in the blank | Foundation | Classic-method variant

Q012 · Writing a Right-Endpoint Sum for a Square

Problem

Partition $[0,1]$ into n equal pieces and use right endpoints. The Riemann sum for $f(x)=x^2$ is ____.

Answer

$\frac{1}{n^3}\sum_{k=1}^{n} k^2$

Knowledge points

- definition by Riemann sums
- subinterval widths and sample points
- limits of sums and definite integrals

Reading and method choice

The width is $\frac{1}{n}$ and the right endpoint is $\frac{k}{n}$; substitute into the square and multiply by the width.

Detailed derivation

1. Each width is $\Delta x = \frac{1}{n}$.
2. The kth right endpoint is $x_k = \frac{k}{n}$.
3. The sampled value is $f(x_k) = \left(\frac{k}{n}\right)^2$.
4. Thus the sum is $\sum_{k=1}^{n} \left(\frac{k}{n}\right)^2 \frac{1}{n} = \frac{1}{n^3}\sum_{k=1}^{n} k^2$.

Common pitfalls

- omitting the width factor Δx
- failing to check both sample points and the integration interval

Verification

Using $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$, the limiting value is $\frac{1}{3}$, agreeing with $\int_0^1 x^2\,dx$.

Method takeaway

A Riemann sum is identified only after matching the function value, sample points, subinterval width, and interval.

Extension

Left endpoints or midpoints may be used instead; continuity gives the same limiting integral.

Source lineage: Classic-method variant · riemann_sums_and_definition · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic Calculation Foundation Classic-method variant

Q013 · Converting a Limit into a Definite Integral

Problem

Evaluate $\lim_{n\to\infty} \frac{27}{n^3} \sum_{k=1}^{n} k^2$ by first identifying it as a definite integral.

Answer

9

Knowledge points

- definition by Riemann sums
- subinterval widths and sample points
- limits of sums and definite integrals

Reading and method choice

Regroup the factors as the sampled value $\left(\frac{3k}{n}\right)^2$ times the width $\frac{3}{n}$.

Detailed derivation

1. Rewrite the sum as $\frac{27}{n^3} \sum_{k=1}^{n} k^2 = \sum_{k=1}^{n} \left(\frac{3k}{n}\right)^2 \frac{3}{n}$.
2. Here $\Delta x = \frac{3}{n}$, $x_k = \frac{3k}{n}$, and the interval is $[0, 3]$.
3. Hence the limit is $\int_0^3 x^2\, dx$.
4. Evaluation gives $\left.\frac{x^3}{3}\right|_0^3 = 9$.

Common pitfalls

- omitting the width factor Δx
- failing to check both sample points and the integration interval

Verification

Direct use of the square-sum formula gives $\frac{27}{n^3} \cdot \frac{n(n+1)(2n+1)}{6} \to 9$, matching the integral method.

Method takeaway

A Riemann sum is identified only after matching the function value, sample points, subinterval width, and interval.

Extension

Left endpoints or midpoints may be used instead; continuity gives the same limiting integral.

Source lineage: Classic-method variant · riemann_sums_and_definition · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic Calculation Foundation Classic-method variant

Q014 · Evaluating an Absolute-Value Integral Piecewise

Problem

Evaluate $\int_0^3 |x-1|\,dx$.

Answer

$\frac{5}{2}$

Knowledge points

- linearity of definite integrals
- interval additivity and reversal
- order bounds and symmetry

Reading and method choice

The formula inside the absolute value changes at $x=1$, so the interval must be split there.

Detailed derivation

1. On $[0,1]$, $|x-1|=1-x$; on $[1,3]$, $|x-1|=x-1$.
2. Thus $\int_0^3 |x-1|\,dx = \int_0^1 (1-x)\,dx + \int_1^3 (x-1)\,dx$.
3. The first term is $x-\frac{x^2}{2}\Big|_0^1 = \frac{1}{2}$.
4. The second is $\frac{(x-1)^2}{2}\Big|_1^3 = 2$, for a total of $\frac{5}{2}$.

Common pitfalls

- identifying a definite integral with unsigned geometric area
- forgetting the sign change when limits are reversed

Verification

Geometrically the two triangles have base-height pairs $1,1$ and $2,2$, so their areas total $\frac{1}{2}+2=\frac{5}{2}$.

Method takeaway

Check sign, interval orientation, and symmetry before carrying out piecewise computation.

Extension

Splitting an integrand into odd and even parts can further simplify integrals over symmetric intervals.

Source lineage: Classic-method variant · definite_integral_properties · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic **Calculation** **Foundation** **Classic-method variant**

Q015 · Combining Integrals over Adjacent Intervals

Problem

Let $a < b < c$, with $\int_a^b f(x)\,dx = 4$ and $\int_b^c f(x)\,dx = -1$. Find $\int_c^a f(x)\,dx$.

Answer

-3

Knowledge points

- linearity of definite integrals
- interval additivity and reversal
- order bounds and symmetry

Reading and method choice

First use interval additivity in the forward direction, then reverse the limits.

Detailed derivation

1. Interval additivity gives $\int_a^c f(x)\,dx = \int_a^b f(x)\,dx + \int_b^c f(x)\,dx$.
2. Substitution of the values yields $\int_a^c f(x)\,dx = 4 + (-1) = 3$.
3. Reversing the limits gives $\int_c^a f(x)\,dx = -\int_a^c f(x)\,dx = -3$.
4. Therefore the requested oppositely oriented integral is -3.

Common pitfalls

- identifying a definite integral with unsigned geometric area
- forgetting the sign change when limits are reversed

Verification

Following the oriented loop gives $\int_a^b f + \int_b^c f + \int_c^a f = 0$; substitution gives $4 - 1 - 3 = 0$.

Method takeaway

Check sign, interval orientation, and symmetry before carrying out piecewise computation.

Extension

Splitting an integrand into odd and even parts can further simplify integrals over symmetric intervals.

Source lineage: Classic-method variant · definite_integral_properties · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic Calculation Foundation Classic-method variant

Q016 · Using Pointwise Bounds to Give the Sharp Guaranteed Range

Problem

Suppose f is integrable on $[-2,2]$ and $-1 \le f(x) \le 3$. Find the range for $\int_{-2}^{2} f(x)\,dx$ guaranteed by this information alone, and show that both bounds are attainable.

Answer

$-4 \le \int_{-2}^{2} f(x)\,dx \le 12$

Knowledge points

- linearity of definite integrals
- interval additivity and reversal
- order bounds and symmetry

Reading and method choice

Integrate the two constant bounds and use constant functions to show that the resulting bounds are the sharp universal guarantee.

Detailed derivation

1. Order preservation gives $\int_{-2}^{2}(-1)\,dx \le \int_{-2}^{2} f(x)\,dx \le \int_{-2}^{2} 3\,dx$.
2. The interval length is 4, so the lower bound is $-1 \cdot 4 = -4$.
3. The upper bound is $3 \cdot 4 = 12$.
4. The choices $f(x) = -1$ and $f(x) = 3$ attain the lower and upper bounds respectively.

Common pitfalls

- identifying a definite integral with unsigned geometric area
- forgetting the sign change when limits are reversed

Verification

Both endpoints occur for functions satisfying the hypothesis, so no narrower universal interval can be guaranteed.

Method takeaway

Check sign, interval orientation, and symmetry before carrying out piecewise computation.

Extension

Splitting an integrand into odd and even parts can further simplify integrals over symmetric intervals.

Source lineage: Classic-method variant · definite_integral_properties · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic Calculation Foundation Classic-method variant

Q017 · Using Symmetry before Evaluation

Problem

Evaluate $\int_{-2}^{2}(x^3+2x^2)\,dx$ and identify the part that vanishes.

Answer

$\frac{32}{3}$

Knowledge points

- linearity of definite integrals
- interval additivity and reversal
- order bounds and symmetry

Reading and method choice

On a symmetric interval the odd part integrates to zero, while the even part becomes twice the half-interval integral.

Detailed derivation

1. x^3 is odd, so $\int_{-2}^{2} x^3\,dx = 0$.
2. $2x^2$ is even, so $\int_{-2}^{2} 2x^2\,dx = 2\int_0^2 2x^2\,dx$.
3. Evaluate the half-interval integral: $2\int_0^2 2x^2\,dx = 4\left.\frac{x^3}{3}\right|_0^2 = \frac{32}{3}$.
4. Thus the original integral is $\frac{32}{3}$, and the x^3 part is the part that vanishes.

Common pitfalls

- identifying a definite integral with unsigned geometric area
- forgetting the sign change when limits are reversed

Verification

Direct evaluation of the antiderivative $\frac{x^4}{4}+\frac{2x^3}{3}$ at -2 and 2 also gives $\frac{32}{3}$.

Method takeaway

Check sign, interval orientation, and symmetry before carrying out piecewise computation.

Extension

Splitting an integrand into odd and even parts can further simplify integrals over symmetric intervals.

Source lineage: Classic-method variant · definite_integral_properties · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic Proof Foundation Original synthesis / diagnosis

Q018 · Proving Interval Additivity from Riemann Sums

Problem

Let f be continuous on $[a, b]$, with $a < c < b$. Starting from Riemann sums, prove $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$.

Answer

See the proof.

Knowledge points

- linearity of definite integrals
- interval additivity and reversal
- order bounds and symmetry

Reading and method choice

Choose partitions containing c, so every whole-interval sum splits exactly into a left sum and a right sum, and then pass to the limits.

Detailed derivation

1. Choose partitions P_1 of $[a, c]$ and P_2 of $[c, b]$, with both mesh sizes tending to 0.
2. Combine their points to obtain a partition $P = P_1 \cup P_2$ of $[a, b]$ that explicitly contains c.
3. With compatible sample points, the sums satisfy the exact identity $S(f, P) = S(f, P_1) + S(f, P_2)$.
4. Continuity guarantees existence of all three Riemann-sum limits. Passing to the limit gives $\int_a^b f = \int_a^c f + \int_c^b f$.

Common pitfalls

- identifying a definite integral with unsigned geometric area
- forgetting the sign change when limits are reversed

Verification

For $f(x) = 1$, the result reduces to the length identity $b - a = (c - a) + (b - c)$, matching the proof structure.

Method takeaway

Check sign, interval orientation, and symmetry before carrying out piecewise computation.

Extension

Splitting an integrand into odd and even parts can further simplify integrals over symmetric intervals.

Source lineage: Original synthesis / diagnosis · definite_integral_properties · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic Proof Foundation Original synthesis / diagnosis

Q019 · Proving the Integral Triangle Inequality

Problem

Let f be continuous on $[a,b]$. Prove $\left|\int_a^b f(x)\,dx\right| \le \int_a^b |f(x)|\,dx$, and state a sufficient condition for equality.

Answer

The inequality holds; equality occurs when f has constant sign on the interval.

Knowledge points

- linearity of definite integrals
- interval additivity and reversal
- order bounds and symmetry

Reading and method choice

First prove the inequality from pointwise two-sided bounds. For a continuous function of constant sign, the absolute value can be removed consistently over the whole interval.

Detailed derivation

1. Pointwise, $-|f(x)| \le f(x) \le |f(x)|$.
2. Order preservation gives $-A \le I \le A$, where $I = \int_a^b f(x)\,dx$ and $A = \int_a^b |f(x)|\,dx \ge 0$.
3. This two-sided inequality is equivalent to $|I| \le A$.
4. If $f(x) \ge 0$, then $|f| = f$; if $f(x) \le 0$, then $|f| = -f$. Either condition gives equality.

Common pitfalls

- identifying a definite integral with unsigned geometric area
- forgetting the sign change when limits are reversed

Verification

For $f(x) = x$ on $[-1,1]$, the left side is 0 and the right side is 1, showing that sign changes can make the inequality strict.

Method takeaway

Check sign, interval orientation, and symmetry before carrying out piecewise computation.

Extension

Splitting an integrand into odd and even parts can further simplify integrals over symmetric intervals.

Source lineage: Original synthesis / diagnosis · definite_integral_properties · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic | Parameter / synthesis / application | Foundation | Original synthesis / diagnosis

Q020 · From a Nonlinear Sum to an Exact Integral

Problem

Find $L = \lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} \sqrt{1+\left(\frac{k}{n}\right)^2}$, identify the corresponding definite integral, and evaluate it exactly.

Answer

$L = \frac{1}{2}\left(\sqrt{2} + \ln(1+\sqrt{2})\right)$

Knowledge points

- definition by Riemann sums
- subinterval widths and sample points
- limits of sums and definite integrals

Reading and method choice

The expression is already a right-endpoint Riemann sum; after identifying it, use the radical antiderivative established in Chapter 4.

Detailed derivation

1. Let $\Delta x = \frac{1}{n}$ and $x_k = \frac{k}{n}$; the sum is $\sum_{k=1}^{n} \sqrt{1+x_k^2}\,\Delta x$.
2. Hence $L = \int_0^1 \sqrt{1+x^2}\,dx$.
3. An antiderivative is $\frac{1}{2}\left(x\sqrt{1+x^2} + \ln\left(x+\sqrt{1+x^2}\right)\right)$.
4. Evaluation at 1 and 0 gives $L = \frac{1}{2}\left(\sqrt{2} + \ln(1+\sqrt{2})\right)$.

Common pitfalls

- omitting the width factor Δx
- failing to check both sample points and the integration interval

Verification

Every summand lies between 1 and $\sqrt{2}$, so the limit must also lie there; the exact answer is approximately 1.1478, which is consistent.

Method takeaway

A Riemann sum is identified only after matching the function value, sample points, subinterval width, and interval.

Extension

Left endpoints or midpoints may be used instead; continuity gives the same limiting integral.

Source lineage: Original synthesis / diagnosis · riemann_sums_and_definition · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic | Single choice | Foundation | Classic-method variant

Q021 · Differentiating a Basic Accumulation Function

Problem

Let $F(x) = \int_0^x (1 + t^4)\, dt$. Which expression equals $F'(x)$?

A. $1 + t^4$

B. $4x^3$

C. $1 + x^4$

D. $x + x^5$

Answer

C

Knowledge points

- accumulation functions
- Fundamental Theorem of Calculus
- chain rule

Reading and method choice

The integrand is continuous and the upper limit is exactly x, so apply Part I of the Fundamental Theorem directly.

Detailed derivation

1. Treat t as a dummy variable internal to the integral.
2. If $F(x) = \int_a^x f(t)\, dt$ with continuous f, then $F'(x) = f(x)$.
3. Here $f(t) = 1 + t^4$, so $F'(x) = 1 + x^4$.
4. Thus C is correct; an explicit formula for $F(x)$ is unnecessary.

Common pitfalls

- omitting the derivative of a variable upper limit
- missing the minus sign from a variable lower limit

Verification

Direct integration gives $F(x) = x + \frac{x^5}{5}$, whose derivative is again $1 + x^4$.

Method takeaway

Apply the theorem at each variable endpoint, multiply by the endpoint derivative, and combine upper minus lower.

Extension

If the integrand also contains the parameter explicitly, rewrite first or apply the product rule rather than looking only at the endpoints.

Source lineage: Classic-method variant · fundamental_theorem_and_new_functions · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic | Single choice | Foundation | Classic-method variant

Q022 · A Variable Lower Limit with a Composite Endpoint

Problem

Let $H(x) = \int_{x^2}^{3} e^{t^2}\,dt$. Which expression equals $H'(x)$?

A. $2xe^{x^2}$

B. $-2xe^{x^4}$

C. e^{x^4}

D. $-e^{x^2}$

Answer

B

Knowledge points

- accumulation functions
- Fundamental Theorem of Calculus
- chain rule

Reading and method choice

The variable appears in the lower limit, producing a minus sign, and the endpoint x^2 contributes its derivative $2x$.

Detailed derivation

1. The general rule is $\frac{d}{dx}\int_{u(x)}^{c} f(t)\,dt = -f(u(x))u'(x)$.
2. Here $u(x) = x^2$, so $u'(x) = 2x$.
3. The endpoint value is $e^{(x^2)^2} = e^{x^4}$.
4. Hence $H'(x) = -2xe^{x^4}$, so B is correct.

Common pitfalls

- omitting the derivative of a variable upper limit
- missing the minus sign from a variable lower limit

Verification

For $x > 0$, the lower limit moves right while the integrand is positive, so the integral should decrease; the answer is indeed negative.

Method takeaway

Apply the theorem at each variable endpoint, multiply by the endpoint derivative, and combine upper minus lower.

Extension

If the integrand also contains the parameter explicitly, rewrite first or apply the product rule rather than looking only at the endpoints.

Source lineage: Classic-method variant · fundamental_theorem_and_new_functions · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | Single choice | Methods | Classic-method variant

Q023 · Evaluating with the Newton-Leibniz Formula

Problem

Which value equals $\int_0^1 3x^2\,dx$?

A. 1

B. $\frac{1}{3}$

C. 3

D. 0

Answer

A

Knowledge points

- Newton-Leibniz formula
- antiderivatives
- exact evaluation of definite integrals

Reading and method choice

Choose the antiderivative x^3 and subtract its lower-end value from its upper-end value.

Detailed derivation

1. An antiderivative of $3x^2$ is $F(x) = x^3$.
2. The Newton-Leibniz formula gives $\int_0^1 3x^2\,dx = F(1) - F(0)$.
3. This is $1^3 - 0^3 = 1$.
4. Therefore A is correct.

Common pitfalls

- substituting only the upper limit
- carrying an arbitrary antiderivative constant into a definite integral

Verification

The integrand is nonnegative on $[0, 1]$, so the result must be nonnegative; the area is 1.

Method takeaway

Choose any antiderivative and subtract its lower-end value from its upper-end value; arbitrary constants cancel.

Extension

After evaluation, use sign, scale, or a geometric interpretation as an independent check.

Source lineage: Classic-method variant · newton_leibniz_evaluation · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | Single choice | Methods | Classic-method variant

Q024 · Recognizing the Newton-Leibniz Formula

Problem

If $F'(x) = f(x)$ and f is continuous on $[a, b]$, which statement is correct?

A. $\int_a^b f(x)\,dx = F(a) - F(b)$

B. $\int_a^b f(x)\,dx = F(b)$

C. $\int_a^b f(x)\,dx = f(b) - f(a)$

D. $\int_a^b f(x)\,dx = F(b) - F(a)$

Answer

D

Knowledge points

- Newton-Leibniz formula
- antiderivatives
- exact evaluation of definite integrals

Reading and method choice

The definite integral equals the upper-end value minus the lower-end value of any antiderivative.

Detailed derivation

1. The hypothesis says that F is an antiderivative of f.
2. The Newton-Leibniz formula is $\int_a^b f(x)\,dx = F(b) - F(a)$.
3. An arbitrary constant cancels from this difference and is not retained.
4. Thus D is correct.

Common pitfalls

- substituting only the upper limit
- carrying an arbitrary antiderivative constant into a definite integral

Verification

After swapping a, b, the right side becomes $F(a) - F(b)$, exactly matching the sign change from reversing limits.

Method takeaway

Choose any antiderivative and subtract its lower-end value from its upper-end value; arbitrary constants cancel.

Extension

After evaluation, use sign, scale, or a geometric interpretation as an independent check.

Source lineage: Classic-method variant · newton_leibniz_evaluation · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | Multiple choice | Methods | Classic-method variant

Q025 · Necessary Properties of an Accumulation Function

Problem

Let f be continuous on $[a, b]$ and $F(x) = \int_a^x f(t)\,dt$. Which conclusions must hold?

A. For $x \in (a, b)$, $F'(x) = f(x)$

B. $F(a) = 0$

C. $F''(x)$ must exist throughout the interval.

D. $F(b) = \int_a^b f(t)\,dt$

Answer

A, B, D

Knowledge points

- accumulation functions
- Fundamental Theorem of Calculus
- chain rule

Reading and method choice

Continuity permits the Fundamental Theorem. Check the initial value, derivative, and endpoint value; a second derivative is not guaranteed.

Detailed derivation

1. The Fundamental Theorem gives $F'(x) = f(x)$ at interior points (with one-sided derivatives at the endpoints), so A is correct.
2. $F(a) = \int_a^a f(t)\,dt = 0$, so B is correct.
3. Existence of F'' would require differentiability of f, which continuity alone does not guarantee; C is false.
4. $F(b) = \int_a^b f(t)\,dt$ follows directly from the definition, so D is correct.

Common pitfalls

- omitting the derivative of a variable upper limit
- missing the minus sign from a variable lower limit

Verification

For example, on $[-1, 1]$ take the continuous function $f(x) = |x|$, which is not differentiable at 0; then $F''(0)$ does not exist, confirming that C is not guaranteed.

Method takeaway

Apply the theorem at each variable endpoint, multiply by the endpoint derivative, and combine upper minus lower.

Extension

If the integrand also contains the parameter explicitly, rewrite first or apply the product rule rather than looking only at the endpoints.

Source lineage: Classic-method variant · fundamental_theorem_and_new_functions · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | Multiple choice | Methods | Classic-method variant

Q026 · Rules for an Integral with Two Variable Limits

Problem

Let f be continuous, let u, v be differentiable, and define $H(x) = \int_{u(x)}^{v(x)} f(t)\,dt$. Which conclusions are correct?

A. $H'(x) = f(v(x))v'(x) - f(u(x))u'(x)$

B. If $u(x) = v(x)$, then $H(x) = 0$.

C. Interchanging u, v leaves H unchanged.

D. If u is constant and $v(x) = x$, then $H'(x) = f(x)$.

Answer

A, B, D

Knowledge points

- accumulation functions
- Fundamental Theorem of Calculus
- chain rule

Reading and method choice

Write the integral as the difference of two accumulation functions with a common base point to handle both limits.

Detailed derivation

1. For a fixed base point c, write $H(x) = \int_c^{v(x)} f(t)\,dt - \int_c^{u(x)} f(t)\,dt$.
2. Differentiating gives $H'(x) = f(v(x))v'(x) - f(u(x))u'(x)$, so A is correct.
3. If $u(x) = v(x)$, the equal-limit integral is 0, so B is correct; swapping limits changes the sign, so C is false.
4. With constant u and $v(x) = x$, A reduces to $H'(x) = f(x)$, so D is correct.

Common pitfalls

- omitting the derivative of a variable upper limit
- missing the minus sign from a variable lower limit

Verification

For $f(t) = 1$, $H(x) = v(x) - u(x)$ and $H'(x) = v'(x) - u'(x)$, exactly matching A.

Method takeaway

Apply the theorem at each variable endpoint, multiply by the endpoint derivative, and combine upper minus lower.

Extension

If the integrand also contains the parameter explicitly, rewrite first or apply the product rule rather than looking only at the endpoints.

Source lineage: Classic-method variant · fundamental_theorem_and_new_functions · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | True/false with justification | Methods | Classic-method variant

Q027 · The Continuity Hypothesis in the Fundamental Theorem

Problem

Determine whether the statement is true and justify: if f is continuous on $[a, b]$ and $F(x) = \int_a^x f(t)\,dt$, then $F'(x) = f(x)$ for every $x \in (a, b)$.

Answer

True.

Knowledge points

- accumulation functions
- Fundamental Theorem of Calculus
- chain rule

Reading and method choice

Write the derivative quotient as the average of the function over a short interval, then use continuity to make that average approach the point value.

Detailed derivation

1. $\frac{F(x+h)-F(x)}{h} = \frac{1}{h}\int_x^{x+h} f(t)\,dt$.
2. The integral mean value theorem supplies a point ξ_h between x and $x+h$ for which the quotient equals $f(\xi_h)$.
3. As $h \to 0$, $\xi_h \to x$.
4. Continuity at x gives $f(\xi_h) \to f(x)$, hence $F'(x) = f(x)$.

Common pitfalls

- omitting the derivative of a variable upper limit
- missing the minus sign from a variable lower limit

Verification

For $f(t) = t^2$, $F(x) = \frac{x^3-a^3}{3}$, whose derivative is $x^2 = f(x)$.

Method takeaway

Apply the theorem at each variable endpoint, multiply by the endpoint derivative, and combine upper minus lower.

Extension

If the integrand also contains the parameter explicitly, rewrite first or apply the product rule rather than looking only at the endpoints.

Source lineage: Classic-method variant · fundamental_theorem_and_new_functions · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | True/false with justification | Methods | Classic-method variant

Q028 · One Zero Accumulation Value Does Not Determine an Endpoint Value

Problem

Determine whether the statement is true and justify: if a continuous function satisfies $\int_0^1 f(t)\,dt = 0$, then necessarily $f(1) = 0$.

Answer

False.

Knowledge points

- accumulation functions
- Fundamental Theorem of Calculus
- chain rule

Reading and method choice

A zero integral may result from cancellation rather than a zero endpoint value; a linear counterexample is enough.

Detailed derivation

1. Take the continuous function $f(t) = 2t - 1$.
2. Then $\int_0^1 (2t-1)\,dt = t^2 - t\big|_0^1 = 0$.
3. However, $f(1) = 2 \cdot 1 - 1 = 1$.
4. The hypothesis holds while the conclusion fails, so the statement is false.

Common pitfalls

- omitting the derivative of a variable upper limit
- missing the minus sign from a variable lower limit

Verification

The function is negative to the left of $\frac{1}{2}$ and positive to the right, and the signed areas cancel exactly.

Method takeaway

Apply the theorem at each variable endpoint, multiply by the endpoint derivative, and combine upper minus lower.

Extension

If the integrand also contains the parameter explicitly, rewrite first or apply the product rule rather than looking only at the endpoints.

Source lineage: Classic-method variant · fundamental_theorem_and_new_functions · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | Fill in the blank | Methods | Classic-method variant

Q029 · A Standard Integral for the Natural Logarithm

Problem

Fill in the blank: $\int_1^e \frac{1}{x}\,dx =$ ____.

Answer

1

Knowledge points

- Newton-Leibniz formula
- antiderivatives
- exact evaluation of definite integrals

Reading and method choice

On the positive axis, $\ln x$ is an antiderivative of $\frac{1}{x}$.

Detailed derivation

1. Choose the antiderivative $F(x) = \ln x$.
2. The Newton-Leibniz formula gives $\int_1^e \frac{1}{x}\,dx = \ln e - \ln 1$.
3. Since $\ln e = 1$ and $\ln 1 = 0$, the result is 1.
4. Because $\frac{1}{x} > 0$ and the interval has positive length, the positive result is consistent with the sign of the integrand.

Common pitfalls

- substituting only the upper limit
- carrying an arbitrary antiderivative constant into a definite integral

Verification

The integrand is positive on $[1, e]$, so the positive answer is consistent; this is also the integral definition of the natural logarithm.

Method takeaway

Choose any antiderivative and subtract its lower-end value from its upper-end value; arbitrary constants cancel.

Extension

After evaluation, use sign, scale, or a geometric interpretation as an independent check.

Source lineage: Classic-method variant · newton_leibniz_evaluation · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | Fill in the blank | Methods | Classic-method variant

Q030 · Differentiation with Trigonometric Endpoints

Problem

Let $F(x) = \int_{\sin x}^{\cos x} e^{t^2}\,dt$. Then $F'(x) =$ ____.

Answer

$-\sin x\, e^{\cos^2 x} - \cos x\, e^{\sin^2 x}$

Knowledge points

- accumulation functions
- Fundamental Theorem of Calculus
- chain rule

Reading and method choice

Handle the upper and lower limits separately: the upper term is multiplied by $-\sin x$, and the lower contribution has an additional overall minus sign.

Detailed derivation

1. Set $v(x) = \cos x$ and $u(x) = \sin x$.
2. The two-limit formula gives $F'(x) = e^{v(x)^2} v'(x) - e^{u(x)^2} u'(x)$.
3. Substitute $v'(x) = -\sin x$ and $u'(x) = \cos x$.
4. This yields $F'(x) = -\sin x\, e^{\cos^2 x} - \cos x\, e^{\sin^2 x}$.

Common pitfalls

- omitting the derivative of a variable upper limit
- missing the minus sign from a variable lower limit

Verification

At $x = 0$, the upper limit has zero first-order motion while the lower limit moves right with speed 1, so $F'(0) = -1$; the formula also gives -1.

Method takeaway

Apply the theorem at each variable endpoint, multiply by the endpoint derivative, and combine upper minus lower.

Extension

If the integrand also contains the parameter explicitly, rewrite first or apply the product rule rather than looking only at the endpoints.

Source lineage: Classic-method variant · fundamental_theorem_and_new_functions · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | Fill in the blank | Methods | Classic-method variant

Q031 · An Accumulation Function inside a Product

Problem

Let $G(x) = x \int_0^x t^2\,dt$. Then $G'(x) =$ ____.

Answer

$\frac{4}{3}x^3$

Knowledge points

- accumulation functions
- Fundamental Theorem of Calculus
- chain rule

Reading and method choice

There is an additional factor x, so differentiating only the integral is insufficient; use the product rule.

Detailed derivation

1. Let $A(x) = \int_0^x t^2\,dt$; then $A'(x) = x^2$.
2. The product rule gives $G'(x) = A(x) + xA'(x)$.
3. Also, $A(x) = \frac{x^3}{3}$.
4. Therefore $G'(x) = \frac{x^3}{3} + x^3 = \frac{4}{3}x^3$.

Common pitfalls

- omitting the derivative of a variable upper limit
- missing the minus sign from a variable lower limit

Verification

Simplifying first gives $G(x) = x \cdot \frac{x^3}{3} = \frac{x^4}{3}$, whose derivative is again $\frac{4}{3}x^3$.

Method takeaway

Apply the theorem at each variable endpoint, multiply by the endpoint derivative, and combine upper minus lower.

Extension

If the integrand also contains the parameter explicitly, rewrite first or apply the product rule rather than looking only at the endpoints.

Source lineage: Classic-method variant · fundamental_theorem_and_new_functions · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | Fill in the blank | Methods | Classic-method variant

Q032 · A Definite Integral of Sine

Problem

Fill in the blank: $\int_0^{\frac{\pi}{2}} \sin x \, dx = \_\_\_\_$.

Answer

1

Knowledge points

- Newton-Leibniz formula
- antiderivatives
- exact evaluation of definite integrals

Reading and method choice

$-\cos x$ is an antiderivative of $\sin x$.

Detailed derivation

1. Choose the antiderivative $F(x) = -\cos x$.
2. Evaluate the endpoints: $\int_0^{\frac{\pi}{2}} \sin x \, dx = -\cos x|_0^{\frac{\pi}{2}}$.
3. The result is $-\cos\frac{\pi}{2} - (-\cos 0) = 0 + 1 = 1$.
4. Since $\sin x \geq 0$ throughout the interval, the positive value 1 has the correct sign.

Common pitfalls

- substituting only the upper limit
- carrying an arbitrary antiderivative constant into a definite integral

Verification

In the first quadrant the sine curve lies between 0 and 1 over a length $\frac{\pi}{2}$; an area of 1 is plausible.

Method takeaway

Choose any antiderivative and subtract its lower-end value from its upper-end value; arbitrary constants cancel.

Extension

After evaluation, use sign, scale, or a geometric interpretation as an independent check.

Source lineage: Classic-method variant · newton_leibniz_evaluation · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard **Fill in the blank** **Methods** **Classic-method variant**

Q033 · Evaluating a Composite-Limit Derivative at a Point

Problem

Let $G(x) = \int_1^{x^2} \ln t \, dt$. Then $G'(1) =$ ____.

Answer

0

Knowledge points

- accumulation functions
- Fundamental Theorem of Calculus
- chain rule

Reading and method choice

Find the general derivative first and then substitute the specified point; the upper limit x^2 must not be treated as x.

Detailed derivation

1. The upper-limit function is $u(x) = x^2$, with $u'(x) = 2x$.
2. The Fundamental Theorem and chain rule give $G'(x) = \ln(x^2) \cdot 2x$.
3. At $x = 1$, this becomes $G'(1) = 2 \ln 1$.
4. Since $\ln 1 = 0$, $G'(1) = 0$.

Common pitfalls

- omitting the derivative of a variable upper limit
- missing the minus sign from a variable lower limit

Verification

Near $x = 1$, the integrand vanishes at the moving endpoint $t = 1$, so the first-order change of the accumulation is indeed zero.

Method takeaway

Apply the theorem at each variable endpoint, multiply by the endpoint derivative, and combine upper minus lower.

Extension

If the integrand also contains the parameter explicitly, rewrite first or apply the product rule rather than looking only at the endpoints.

Source lineage: Classic-method variant · fundamental_theorem_and_new_functions · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Calculation Methods Classic-method variant

Q034 · A Definite Integral of a Linear Function

Problem

Evaluate $\int_{-1}^{2}(2x+3)\,dx$.

Answer

12

Knowledge points

- Newton-Leibniz formula
- antiderivatives
- exact evaluation of definite integrals

Reading and method choice

Find a polynomial antiderivative and evaluate it as upper endpoint minus lower endpoint.

Detailed derivation

1. An antiderivative of $2x+3$ is $F(x)=x^2+3x$.
2. The upper-end value is $F(2)=4+6=10$.
3. The lower-end value is $F(-1)=1-3=-2$.
4. Thus the integral is $F(2)-F(-1)=10-(-2)=12$.

Common pitfalls

- substituting only the upper limit
- carrying an arbitrary antiderivative constant into a definite integral

Verification

The endpoint values are 1 and 7, so the average height is 4 over length 3; the trapezoid area is $4\cdot 3=12$.

Method takeaway

Choose any antiderivative and subtract its lower-end value from its upper-end value; arbitrary constants cancel.

Extension

After evaluation, use sign, scale, or a geometric interpretation as an independent check.

Source lineage: Classic-method variant · newton_leibniz_evaluation · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard **Calculation** **Methods** **Classic-method variant**

Q035 · A Definite Integral with the Arctangent Antiderivative

Problem

Evaluate $\int_0^1 \frac{1}{1+x^2}\,dx$.

Answer

$\frac{\pi}{4}$

Knowledge points

- Newton-Leibniz formula
- antiderivatives
- exact evaluation of definite integrals

Reading and method choice

Recognize the standard antiderivative arctan x and evaluate at the endpoints.

Detailed derivation

1. $\frac{d}{dx}$ arctan $x = \frac{1}{1+x^2}$.
2. Hence $\int_0^1 \frac{1}{1+x^2}\,dx =$ arctan $x\big|_0^1$.
3. arctan $1 = \frac{\pi}{4}$ and arctan $0 = 0$.
4. Therefore the value is $\frac{\pi}{4}$.

Common pitfalls

- substituting only the upper limit
- carrying an arbitrary antiderivative constant into a definite integral

Verification

The integrand lies between $\frac{1}{2}$ and 1 on $[0, 1]$, so the integral should lie between those values; $\frac{\pi}{4}$ does.

Method takeaway

Choose any antiderivative and subtract its lower-end value from its upper-end value; arbitrary constants cancel.

Extension

After evaluation, use sign, scale, or a geometric interpretation as an independent check.

Source lineage: Classic-method variant · newton_leibniz_evaluation · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard **Calculation** **Methods** **Classic-method variant**

Q036 · An Exponential Integral over a Finite Interval

Problem

Evaluate $\int_0^{\ln 2} e^{2x}\,dx$.

Answer

$\frac{3}{2}$

Knowledge points

- Newton-Leibniz formula
- antiderivatives
- exact evaluation of definite integrals

Reading and method choice

Differentiating the exponential produces a factor 2, so the antiderivative needs a factor $\frac{1}{2}$.

Detailed derivation

1. An antiderivative of e^{2x} is $F(x) = \frac{1}{2}e^{2x}$.
2. Thus $\int_0^{\ln 2} e^{2x}\,dx = \frac{1}{2}\left(e^{2\ln 2} - e^0\right)$.
3. $e^{2\ln 2} = e^{\ln 4} = 4$ and $e^0 = 1$.
4. Therefore the value is $\frac{1}{2}(4-1) = \frac{3}{2}$.

Common pitfalls

- substituting only the upper limit
- carrying an arbitrary antiderivative constant into a definite integral

Verification

The integrand grows from 1 to 4 over a length of about 0.693; the value 1.5 has the expected scale.

Method takeaway

Choose any antiderivative and subtract its lower-end value from its upper-end value; arbitrary constants cancel.

Extension

After evaluation, use sign, scale, or a geometric interpretation as an independent check.

Source lineage: Classic-method variant · newton_leibniz_evaluation · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Calculation Methods Classic-method variant

Q037 · Integrating the Square of Cosine over a Half Period

Problem

Evaluate $\int_0^{\pi} \cos^2 x \, dx$.

Answer

$\frac{\pi}{2}$

Knowledge points

- Newton-Leibniz formula
- antiderivatives
- exact evaluation of definite integrals

Reading and method choice

Use the power-reduction identity to turn the square into a constant term and a double-angle cosine.

Detailed derivation

1. Use $\cos^2 x = \frac{1+\cos 2x}{2}$.
2. The integral becomes $\frac{1}{2}\int_0^{\pi} 1 \, dx + \frac{1}{2}\int_0^{\pi} \cos 2x \, dx$.
3. The first term is $\frac{\pi}{2}$, and the second is $\frac{1}{4}\sin 2x\Big|_0^{\pi} = 0$.
4. Hence the value is $\frac{\pi}{2}$.

Common pitfalls

- substituting only the upper limit
- carrying an arbitrary antiderivative constant into a definite integral

Verification

The average of $\cos^2 x$ over its full period of length π is $\frac{1}{2}$, so the integral is $\pi \cdot \frac{1}{2}$.

Method takeaway

Choose any antiderivative and subtract its lower-end value from its upper-end value; arbitrary constants cancel.

Extension

After evaluation, use sign, scale, or a geometric interpretation as an independent check.

Source lineage: Classic-method variant · newton_leibniz_evaluation · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Calculation Methods Classic-method variant

Q038 · Differentiating Two Composite Variable Endpoints

Problem

Let $F(x) = \int_x^{x^2} \ln(1+t^2)\,dt$. Find $F'(x)$.

Answer

$F'(x) = 2x\ln(1+x^4) - \ln(1+x^2)$

Knowledge points

- accumulation functions
- Fundamental Theorem of Calculus
- chain rule

Reading and method choice

Both the upper limit x^2 and lower limit x vary, so two terms are required; the upper term also carries the factor $2x$.

Detailed derivation

1. Let $g(t) = \ln(1+t^2)$, $v(x) = x^2$, and $u(x) = x$.
2. The two-limit rule is $F'(x) = g(v(x))v'(x) - g(u(x))u'(x)$.
3. The upper contribution is $\ln(1+x^4) \cdot 2x$.
4. The lower contribution is $\ln(1+x^2) \cdot 1$, so $F'(x) = 2x\ln(1+x^4) - \ln(1+x^2)$.

Common pitfalls

- omitting the derivative of a variable upper limit
- missing the minus sign from a variable lower limit

Verification

At $x = 1$ the endpoints coincide but move at different speeds; the formula gives $F'(1) = \ln 2$, consistent with first-order interval-length speed $2 - 1 = 1$.

Method takeaway

Apply the theorem at each variable endpoint, multiply by the endpoint derivative, and combine upper minus lower.

Extension

If the integrand also contains the parameter explicitly, rewrite first or apply the product rule rather than looking only at the endpoints.

Source lineage: Classic-method variant · fundamental_theorem_and_new_functions · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | Calculation | Methods | Original synthesis / diagnosis

Q039 · An Integral Function with a Parameter-Dependent Kernel

Problem

Let f be continuous and $G(x)=\int_0^x (x-t)f(t)\,dt$. Find $G'(x)$ after first separating the explicit occurrence of x in the integrand.

Answer

$G'(x)=\int_0^x f(t)\,dt$

Knowledge points

- accumulation functions
- Fundamental Theorem of Calculus
- chain rule

Reading and method choice

The Fundamental Theorem cannot be applied mechanically to the whole integrand because it also contains the parameter x explicitly; rewrite by linearity first.

Detailed derivation

1. Expand and separate the parameter: $G(x)=x\int_0^x f(t)\,dt-\int_0^x tf(t)\,dt$.
2. The derivative of the first term by the product rule is $\int_0^x f(t)\,dt+xf(x)$.
3. The derivative of the second term by the Fundamental Theorem is $xf(x)$.
4. The two $xf(x)$ terms cancel, leaving $G'(x)=\int_0^x f(t)\,dt$.

Common pitfalls

- omitting the derivative of a variable upper limit
- missing the minus sign from a variable lower limit

Verification

For $f(t)=1$, $G(x)=\int_0^x (x-t)\,dt=\frac{x^2}{2}$, whose derivative is $x=\int_0^x 1\,dt$.

Method takeaway

Apply the theorem at each variable endpoint, multiply by the endpoint derivative, and combine upper minus lower.

Extension

If the integrand also contains the parameter explicitly, rewrite first or apply the product rule rather than looking only at the endpoints.

Source lineage: Original synthesis / diagnosis · fundamental_theorem_and_new_functions · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | Calculation | Methods | Original synthesis / diagnosis

Q040 · Recovering an Integrand from Its Accumulation

Problem

A continuous function f satisfies $\int_0^x f(t)\,dt = x^2e^x$. Find $f(x)$.

Answer

$f(x) = e^x(x^2 + 2x)$

Knowledge points

- accumulation functions
- Fundamental Theorem of Calculus
- chain rule

Reading and method choice

Differentiate both sides with respect to x. The Fundamental Theorem recovers $f(x)$ on the left, while the product rule handles the right.

Detailed derivation

1. Continuity gives $\frac{d}{dx}\int_0^x f(t)\,dt = f(x)$.
2. Differentiate the right side: $\frac{d}{dx}(x^2e^x) = 2xe^x + x^2e^x$.
3. Factoring gives $2xe^x + x^2e^x = e^x(x^2 + 2x)$.
4. Therefore $f(x) = e^x(x^2 + 2x)$.

Common pitfalls

- omitting the derivative of a variable upper limit
- missing the minus sign from a variable lower limit

Verification

Integrating the answer gives $\int_0^x e^t(t^2 + 2t)\,dt = t^2e^t\big|_0^x = x^2e^x$, recovering the given identity.

Method takeaway

Apply the theorem at each variable endpoint, multiply by the endpoint derivative, and combine upper minus lower.

Extension

If the integrand also contains the parameter explicitly, rewrite first or apply the product rule rather than looking only at the endpoints.

Source lineage: Original synthesis / diagnosis · fundamental_theorem_and_new_functions · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Calculation Methods Original synthesis / diagnosis

Q041 · A Second Derivative of an Accumulation Function

Problem

Let $F(x) = \int_0^{x^2} \cos t \, dt$. Find $F''(x)$.

Answer

$F''(x) = 2\cos(x^2) - 4x^2 \sin(x^2)$

Knowledge points

- accumulation functions
- Fundamental Theorem of Calculus
- chain rule

Reading and method choice

First use the Fundamental Theorem and chain rule for the first derivative, then differentiate the product $2x\cos(x^2)$.

Detailed derivation

1. The first derivative is $F'(x) = \cos(x^2) \cdot 2x = 2x\cos(x^2)$.
2. Apply the product rule to $2x\cos(x^2)$.
3. The first contribution is $2\cos(x^2)$.
4. The second is $2x \cdot [-\sin(x^2)] \cdot 2x = -4x^2 \sin(x^2)$, giving the stated result.

Common pitfalls

- omitting the derivative of a variable upper limit
- missing the minus sign from a variable lower limit

Verification

Since $F(x) = \sin(x^2)$, differentiating this explicit expression twice gives the same result.

Method takeaway

Apply the theorem at each variable endpoint, multiply by the endpoint derivative, and combine upper minus lower.

Extension

If the integrand also contains the parameter explicitly, rewrite first or apply the product rule rather than looking only at the endpoints.

Source lineage: Original synthesis / diagnosis · fundamental_theorem_and_new_functions · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Proof Methods Original synthesis / diagnosis

Q042 · Proving the Fundamental Theorem with the Integral Mean Value Theorem

Problem

Let f be continuous on $[a,b]$ and $F(x)=\int_a^x f(t)\,dt$. Prove that $F'(x)=f(x)$ for $x\in(a,b)$.

Answer

See the proof.

Knowledge points

- accumulation functions
- Fundamental Theorem of Calculus
- chain rule

Reading and method choice

The key is to turn the difference quotient into an average over a shrinking interval and control its mean-value point using continuity.

Detailed derivation

1. For sufficiently small nonzero h, $F(x+h)-F(x)=\int_x^{x+h} f(t)\,dt$.
2. Thus $\frac{F(x+h)-F(x)}{h}=\frac{1}{h}\int_x^{x+h} f(t)\,dt$; oriented integrals make this valid for both signs of h.
3. The integral mean value theorem gives a point ξ_h between x and $x+h$ such that the quotient equals $f(\xi_h)$.
4. As $h\to 0$, $\xi_h\to x$; continuity yields $f(\xi_h)\to f(x)$, so the difference quotient tends to $f(x)$.

Common pitfalls

- omitting the derivative of a variable upper limit
- missing the minus sign from a variable lower limit

Verification

For a constant function, the quotient is identically that constant, consistent with the mean-value-point mechanism in the proof.

Method takeaway

Apply the theorem at each variable endpoint, multiply by the endpoint derivative, and combine upper minus lower.

Extension

If the integrand also contains the parameter explicitly, rewrite first or apply the product rule rather than looking only at the endpoints.

Source lineage: Original synthesis / diagnosis · fundamental_theorem_and_new_functions · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Proof Methods Original synthesis / diagnosis

Q043 · A Zero Accumulation Function Forces a Zero Integrand

Problem

Let f be continuous on $[a, b]$, and suppose $\int_a^x f(t)\,dt = 0$ for every $x \in [a, b]$. Prove that $f(x) = 0$ throughout $[a, b]$.

Answer

See the proof.

Knowledge points

- accumulation functions
- Fundamental Theorem of Calculus
- chain rule

Reading and method choice

The hypothesis is stronger than in the earlier counterexample: the accumulation vanishes for every upper limit, so the identity may be differentiated.

Detailed derivation

1. Define $F(x) = \int_a^x f(t)\,dt$.
2. The hypothesis says $F(x) \equiv 0$, so $F'(x) = 0$ at every interior point.
3. By the Fundamental Theorem and continuity of f, $F'(x) = f(x)$.
4. Hence $f(x) = 0$ in the interior; continuity then gives zero values at the endpoints as well.

Common pitfalls

- omitting the derivative of a variable upper limit
- missing the minus sign from a variable lower limit

Verification

Conversely, if $f \equiv 0$, every accumulation integral is 0, so the conclusion is fully consistent with the hypothesis.

Method takeaway

Apply the theorem at each variable endpoint, multiply by the endpoint derivative, and combine upper minus lower.

Extension

If the integrand also contains the parameter explicitly, rewrite first or apply the product rule rather than looking only at the endpoints.

Source lineage: Original synthesis / diagnosis · fundamental_theorem_and_new_functions · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | Error diagnosis | Methods | Original synthesis / diagnosis

Q044 · Diagnosing Missing Factors in Two-Limit Differentiation

Problem

For $H(x) = \int_x^{x^2} f(t)\,dt$, a student writes $H'(x) = f(x^2) - f(x)$. Identify the error and give the correct result when f is continuous.

Answer

The derivative of the upper limit x^2 was omitted; $H'(x) = 2xf(x^2) - f(x)$.

Knowledge points

- accumulation functions
- Fundamental Theorem of Calculus
- chain rule

Reading and method choice

The student remembered upper minus lower but failed to apply the chain rule to the composite endpoint.

Detailed derivation

1. Set the upper limit $v(x) = x^2$ and lower limit $u(x) = x$.
2. The correct rule is $H'(x) = f(v(x))v'(x) - f(u(x))u'(x)$.
3. Here $v'(x) = 2x$ and $u'(x) = 1$.
4. Therefore $H'(x) = 2xf(x^2) - f(x)$; the student's answer is missing the factor $2x$.

Common pitfalls

- omitting the derivative of a variable upper limit
- missing the minus sign from a variable lower limit

Verification

With $f(t) = 1$, $H(x) = x^2 - x$ and $H'(x) = 2x - 1$, whereas the student's formula gives 0, exposing the error immediately.

Method takeaway

Apply the theorem at each variable endpoint, multiply by the endpoint derivative, and combine upper minus lower.

Extension

If the integrand also contains the parameter explicitly, rewrite first or apply the product rule rather than looking only at the endpoints.

Source lineage: Original synthesis / diagnosis · fundamental_theorem_and_new_functions · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Calculation Methods Classic-method variant

Q045 · Substitution in a Logarithmic Definite Integral

Problem

Use substitution to evaluate $\int_0^1 \frac{2x}{1+x^2}\,dx$, transforming both limits.

Answer

$\ln 2$

Knowledge points

- substitution in definite integrals
- matching differentials
- transforming both limits

Reading and method choice

The derivative of the denominator $1+x^2$ is exactly the numerator $2x$, so use it as the new variable.

Detailed derivation

1. Let $u=1+x^2$, so $du=2x\,dx$.
2. When $x=0$, $u=1$; when $x=1$, $u=2$.
3. The integral becomes $\int_1^2 \frac{1}{u}\,du$.
4. Thus its value is $\ln u\big|_1^2=\ln 2-\ln 1=\ln 2$.

Common pitfalls

- keeping the old limits after changing variables
- losing a constant factor from the differential

Verification

The original integrand is $\frac{d}{dx}\ln(1+x^2)$, so direct endpoint evaluation also gives $\ln 2$.

Method takeaway

The variable, differential, and both limits must be transformed together, leaving only the new variable.

Extension

One may instead keep the old limits, return to the original variable after antidifferentiation, and obtain the same result.

Source lineage: Classic-method variant · definite_integral_substitution · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced | Calculation | Methods | Original synthesis / diagnosis

Q046 · A Trigonometric Substitution with Reversed New Limits

Problem

Use substitution to evaluate $\int_0^{\frac{\pi}{2}} \sin x \cos^3 x\,dx$.

Answer

$\frac{1}{4}$

Knowledge points

- substitution in definite integrals
- matching differentials
- transforming both limits

Reading and method choice

With $u = \cos x$, $du = -\sin x\,dx$ and the new limits run from 1 to 0; handle the minus sign together with the reversed orientation.

Detailed derivation

1. Let $u = \cos x$, so $du = -\sin x\,dx$.
2. When $x = 0$, $u = 1$; when $x = \frac{\pi}{2}$, $u = 0$.
3. The integral becomes $-\int_1^0 u^3\,du = \int_0^1 u^3\,du$.
4. Evaluation gives $\frac{u^4}{4}\Big|_0^1 = \frac{1}{4}$.

Common pitfalls

- keeping the old limits after changing variables
- losing a constant factor from the differential

Verification

The original integrand is nonnegative and at most 1, so the result must lie between 0 and $\frac{\pi}{2}$; $\frac{1}{4}$ has the correct sign and scale.

Method takeaway

The variable, differential, and both limits must be transformed together, leaving only the new variable.

Extension

One may instead keep the old limits, return to the original variable after antidifferentiation, and obtain the same result.

Source lineage: Original synthesis / diagnosis · definite_integral_substitution · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Calculation Methods Classic-method variant

Q047 · Integration by Parts for a Polynomial Times an Exponential

Problem

Use integration by parts to evaluate $\int_0^1 xe^x\,dx$.

Answer

1

Knowledge points

- integration by parts for definite integrals
- boundary terms
- choice of factors

Reading and method choice

Differentiate the polynomial factor and integrate the exponential factor so that the remaining integral is simpler.

Detailed derivation

1. Choose $u = x$ and $dv = e^x\,dx$, so $du = dx$ and $v = e^x$.
2. Definite integration by parts gives $\int_0^1 xe^x\,dx = xe^x|_0^1 - \int_0^1 e^x\,dx$.
3. The boundary term is e, and the remaining integral is $e - 1$.
4. Hence the result is $e - (e - 1) = 1$.

Common pitfalls

- omitting the boundary term
- forgetting to subtract the remaining integral after evaluating the boundary

Verification

An antiderivative is $(x - 1)e^x$; evaluation at 1 and 0 gives $0 - (-1) = 1$.

Method takeaway

Definite integration by parts requires both the evaluated product term and the remaining integral.

Extension

Compare the complexity after swapping the factor choices to assess whether the original selection was efficient.

Source lineage: Classic-method variant · definite_integral_by_parts · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced | Calculation | Methods | Original synthesis / diagnosis

Q048 · Integration by Parts with a Trigonometric Factor

Problem

Evaluate $\int_0^{\frac{\pi}{2}} x \sin x \, dx$.

Answer

1

Knowledge points

- integration by parts for definite integrals
- boundary terms
- choice of factors

Reading and method choice

Differentiate x and integrate $\sin x$, paying close attention to the sign in $v = -\cos x$.

Detailed derivation

1. Choose $u = x$ and $dv = \sin x \, dx$, so $du = dx$ and $v = -\cos x$.
2. Integration by parts gives $\int_0^{\frac{\pi}{2}} x \sin x \, dx = -x \cos x|_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} \cos x \, dx$.
3. The boundary term is 0 at both endpoints.
4. The remaining integral is $\sin x|_0^{\frac{\pi}{2}} = 1$.

Common pitfalls

- omitting the boundary term
- forgetting to subtract the remaining integral after evaluating the boundary

Verification

The integrand is nonnegative on the interval, so the answer must be positive; numerical integration also gives approximately 1.0000.

Method takeaway

Definite integration by parts requires both the evaluated product term and the remaining integral.

Extension

Compare the complexity after swapping the factor choices to assess whether the original selection was efficient.

Source lineage: Original synthesis / diagnosis · definite_integral_by_parts · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced | Proof | Methods | Original synthesis / diagnosis

Q049 · An Interval-Reflection Substitution Identity

Problem

Let f be continuous on $[0, a]$ with $a > 0$. Prove $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$.

Answer

See the proof.

Knowledge points

- substitution in definite integrals
- matching differentials
- transforming both limits

Reading and method choice

Use the reflection substitution $u = a - x$; the new limits reverse while the differential contributes a minus sign, and the two effects cancel.

Detailed derivation

1. In the right-hand integral let $u = a - x$, so $du = -dx$.
2. When $x = 0$, $u = a$; when $x = a$, $u = 0$.
3. Thus $\int_0^a f(a-x)\,dx = -\int_a^0 f(u)\,du$.
4. Reversing the limits gives $\int_0^a f(u)\,du$, which is the left side after renaming the dummy variable.

Common pitfalls

- keeping the old limits after changing variables
- losing a constant factor from the differential

Verification

For $f(x) = x$, the left side is $\frac{a^2}{2}$ and the right side $\int_0^a (a-x)\,dx$ is also $\frac{a^2}{2}$.

Method takeaway

The variable, differential, and both limits must be transformed together, leaving only the new variable.

Extension

One may instead keep the old limits, return to the original variable after antidifferentiation, and obtain the same result.

Source lineage: Original synthesis / diagnosis · definite_integral_substitution · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Proof Methods Original synthesis / diagnosis

Q050 · A Symmetric Ratio Integral via Complementary Angles

Problem

Let f be continuous and strictly positive on $[0,1]$. Prove $\int_0^{\frac{\pi}{2}} \frac{f(\sin x)}{f(\sin x)+f(\cos x)}\,dx = \frac{\pi}{4}$.

Answer

See the proof.

Knowledge points

- substitution in definite integrals
- matching differentials
- transforming both limits

Reading and method choice

Call the integral I. A complementary-angle substitution swaps sine and cosine in the numerator; add the original and transformed forms.

Detailed derivation

1. Let $I = \int_0^{\frac{\pi}{2}} \frac{f(\sin x)}{f(\sin x)+f(\cos x)}\,dx$. Strict positivity ensures the denominator never vanishes.
2. Set $u = \frac{\pi}{2} - x$ and use $\sin\left(\frac{\pi}{2} - u\right) = \cos u$ and $\cos\left(\frac{\pi}{2} - u\right) = \sin u$.
3. This gives $I = \int_0^{\frac{\pi}{2}} \frac{f(\cos u)}{f(\cos u)+f(\sin u)}\,du$.
4. Adding the two representations yields $2I = \int_0^{\frac{\pi}{2}} 1\,dx = \frac{\pi}{2}$, hence $I = \frac{\pi}{4}$.

Common pitfalls

- keeping the old limits after changing variables
- losing a constant factor from the differential

Verification

For $f(s) = 1$, the integrand is identically $\frac{1}{2}$, and the integral is immediately $\frac{1}{2} \cdot \frac{\pi}{2} = \frac{\pi}{4}$.

Method takeaway

The variable, differential, and both limits must be transformed together, leaving only the new variable.

Extension

One may instead keep the old limits, return to the original variable after antidifferentiation, and obtain the same result.

Source lineage: Original synthesis / diagnosis · definite_integral_substitution · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Fill in the blank Synthesis Open-text method adaptation

Q051 · A Composite-Polynomial Substitution

Problem

Evaluate $\int_0^1 x(1+x^2)^2\,dx =$ ___.

Answer

$\frac{7}{6}$

Knowledge points

- substitution in definite integrals
- transforming the variable, differential, and limits together
- orientation of transformed limits

Reading and method choice

The differential of the inner expression $1+x^2$ differs from the factor $x\,dx$ only by a constant, leaving a directly integrable square.

Detailed derivation

1. Set $u = 1+x^2$, so $du = 2x\,dx$ and $x\,dx = \frac{1}{2}du$.
2. When $x = 0$, $u = 1$; when $x = 1$, $u = 2$.
3. The integral becomes $\frac{1}{2}\int_1^2 u^2\,du$.
4. Therefore the value is $\frac{1}{2}\left.\frac{u^3}{3}\right|_1^2 = \frac{1}{6}(8-1) = \frac{7}{6}$.

Common pitfalls

- mixing a new variable with old limits
- changing limits through a non-one-to-one map without splitting
- losing a constant or minus sign in the differential

Verification

Expanding gives $x(1+x^2)^2 = x + 2x^3 + x^5$; termwise integration gives $\frac{1}{2} + 2\cdot\frac{1}{4} + \frac{1}{6} = \frac{7}{6}$, confirming the substitution.

Method takeaway

A definite-integral substitution must transform the variable, differential, and endpoints as one coherent system; the substitution determines orientation.

Extension

Back-substitute before evaluating at the original endpoints as an independent verification route.

Source lineage: Open-text method adaptation · definite_integral_substitution · reference IDs: mit-18.01sc-definite-integral, openstax-calculus-v1-5.5. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | Single choice | Synthesis | Open-text method adaptation

Q052 · A Definite Integral of an Exponential Composite

Problem

Which value equals $\int_0^1 xe^{x^2}\,dx$?

A. $e-1$

B. $\frac{e-1}{2}$

C. $\frac{e}{2}$

D. $\frac{1}{2}$

Answer

B

Knowledge points

- substitution in definite integrals
- transforming the variable, differential, and limits together
- orientation of transformed limits

Reading and method choice

With $u = x^2$, the differential contributes the factor $\frac{1}{2}$, and the new endpoints remain 0 and 1.

Detailed derivation

1. Let $u = x^2$, so $du = 2x\,dx$ and $x\,dx = \frac{1}{2}du$.
2. The endpoints become $x = 0 \Rightarrow u = 0$ and $x = 1 \Rightarrow u = 1$.
3. Thus the integral is $\frac{1}{2}\int_0^1 e^u\,du$.
4. Its value is $\frac{1}{2}(e-1)$, so B is correct.

Common pitfalls

- mixing a new variable with old limits
- changing limits through a non-one-to-one map without splitting

- losing a constant or minus sign in the differential

Verification

Since $xe^{x^2} < e$ on an interval of length 1, the result must be below e; $\frac{e-1}{2}$ satisfies this bound.

Method takeaway

A definite-integral substitution must transform the variable, differential, and endpoints as one coherent system; the substitution determines orientation.

Extension

Back-substitute before evaluating at the original endpoints as an independent verification route.

Source lineage: Open-text method adaptation · definite_integral_substitution · reference IDs: mit-18.01sc-definite-integral, openstax-calculus-v1-5.5. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | Multiple choice | Synthesis | Original synthesis / diagnosis

Q053 · Consistency of Substitution and Endpoints

Problem

Which substitution results are fully correct? Assume every displayed f is continuous.

A. $\displaystyle\int_0^1 2xf(x^2)\,dx = \int_0^1 f(u)\,du$

B. $\displaystyle\int_1^e \frac{f(\ln x)}{x}\,dx = \int_0^1 f(u)\,du$

C. With $u = \cos x$, $\displaystyle\int_0^{\frac{\pi}{2}} \sin x\,dx = \int_1^0 du$

D. If $a > 0$, $\displaystyle\int_0^a f\left(\frac{x}{a}\right)\,dx = a\int_0^1 f(u)\,du$

Answer

A, B, and D.

Knowledge points

- substitution in definite integrals
- transforming the variable, differential, and limits together
- orientation of transformed limits

Reading and method choice

For each choice, check the differential factor, endpoints, and variable consistency; accidental numerical equality cannot replace a valid substitution.

Detailed derivation

1. In A, $u = x^2$ gives $du = 2x\,dx$ and leaves the endpoints $0, 1$, so A is correct.
2. In B, $u = \ln x$ gives $du = \frac{dx}{x}$ and sends $1, e$ to $0, 1$, so B is correct.
3. In C, $u = \cos x$ gives $du = -\sin x\,dx$; retaining endpoints $1, 0$ requires a leading minus sign, so C is false.
4. In D, $u = \frac{x}{a}$ gives $dx = a\,du$; for $a > 0$, endpoints $0, a$ become $0, 1$, so D is correct.

Common pitfalls

- mixing a new variable with old limits
- changing limits through a non-one-to-one map without splitting
- losing a constant or minus sign in the differential

Verification

For C, direct evaluation gives 1 on the left but $\int_1^0 du = -1$ on the displayed right, exposing the sign error.

Method takeaway

A definite-integral substitution must transform the variable, differential, and endpoints as one coherent system; the substitution determines orientation.

Extension

Back-substitute before evaluating at the original endpoints as an independent verification route.

Source lineage: Original synthesis / diagnosis · definite_integral_substitution · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | True/false with justification | Synthesis | Original synthesis / diagnosis

Q054 · Endpoint Orientation under a Decreasing Substitution

Problem

True or false: if $u = \varphi(x)$ is strictly decreasing on $[a, b]$, one may always put the smaller transformed endpoint below the larger one without adding a minus sign.

Answer

False. Reversing the transformed limits introduces a minus sign.

Knowledge points

- substitution in definite integrals
- transforming the variable, differential, and limits together
- orientation of transformed limits

Reading and method choice

The valid formula first keeps the oriented endpoints from $\varphi(a)$ to $\varphi(b)$; manually reordering them requires a compensating minus sign.

Detailed derivation

1. The substitution formula is $\int_a^b f(\varphi(x))\varphi'(x)\,dx = \int_{\varphi(a)}^{\varphi(b)} f(u)\,du$.
2. Strict decrease gives $\varphi(a) > \varphi(b)$, so the transformed integral is naturally reversed.
3. Rewriting it from $\varphi(b)$ to $\varphi(a)$ requires multiplication by -1.
4. For example, $u = 1 - x$ sends $[0, 1]$ to $[1, 0]$; omitting the sign changes the constant-function integral from -1 to 1.

Common pitfalls

- mixing a new variable with old limits
- changing limits through a non-one-to-one map without splitting
- losing a constant or minus sign in the differential

Verification

Take $f(u) = 1$ and $\varphi(x) = 1 - x$: the left side is $\int_0^1 (-1)\,dx = -1$, whereas the reordered integral without a minus sign is 1.

Method takeaway

A definite-integral substitution must transform the variable, differential, and endpoints as one coherent system; the substitution determines orientation.

Extension

Back-substitute before evaluating at the original endpoints as an independent verification route.

Source lineage: Original synthesis / diagnosis · definite_integral_substitution · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard **Calculation** **Synthesis** **Open-text method adaptation**

Q055 · Substitution for an Odd Power of Sine

Problem

Evaluate $\int_0^{\frac{\pi}{2}} \sin^3 x \cos^2 x \, dx$.

Answer

$\frac{2}{15}$

Knowledge points

- substitution in definite integrals
- transforming the variable, differential, and limits together
- orientation of transformed limits

Reading and method choice

Reserve one factor $\sin x \, dx$, rewrite the remaining $\sin^2 x$ as $1 - \cos^2 x$, and set $u = \cos x$.

Detailed derivation

1. Rewrite $\sin^3 x \cos^2 x = (1 - \cos^2 x) \cos^2 x \sin x$.
2. Set $u = \cos x$, so $du = -\sin x \, dx$.
3. The endpoints $x = 0, \frac{\pi}{2}$ become $u = 1, 0$, hence the integral is $\int_0^1 (u^2 - u^4) \, du$.
4. Evaluation gives $\left(\frac{u^3}{3} - \frac{u^5}{5}\right)\Big|_0^1 = \frac{1}{3} - \frac{1}{5} = \frac{2}{15}$.

Common pitfalls

- mixing a new variable with old limits
- changing limits through a non-one-to-one map without splitting
- losing a constant or minus sign in the differential

Verification

The integrand is nonnegative and at most 1, so the value lies between 0 and $\frac{\pi}{2}$; $\frac{2}{15}$ satisfies this and agrees with numerical evaluation.

Method takeaway

A definite-integral substitution must transform the variable, differential, and endpoints as one coherent system; the substitution determines orientation.

Extension

Back-substitute before evaluating at the original endpoints as an independent verification route.

Source lineage: Open-text method adaptation · definite_integral_substitution · reference IDs: mit-18.01sc-definite-integral, openstax-calculus-v1-5.5. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | Calculation | Synthesis | Open-text method adaptation

Q056 · A Reverse-Chain Structure in a Radical

Problem

Evaluate $\int_0^{\frac{\sqrt{3}}{2}} \frac{x}{\sqrt{1-x^2}}\,dx$.

Answer

$\frac{1}{2}$

Knowledge points

- substitution in definite integrals
- transforming the variable, differential, and limits together
- orientation of transformed limits

Reading and method choice

The differential of $1-x^2$ is $-2x\,dx$; the transformed endpoints decrease from 1 to $\frac{1}{4}$.

Detailed derivation

1. Set $u = 1-x^2$, so $du = -2x\,dx$.
2. When $x = 0$, $u = 1$; when $x = \frac{\sqrt{3}}{2}$, $u = \frac{1}{4}$.
3. The integral becomes $-\frac{1}{2}\int_1^{\frac{1}{4}} u^{-\frac{1}{2}}\,du = \frac{1}{2}\int_{\frac{1}{4}}^{1} u^{-\frac{1}{2}}\,du$.
4. Thus the value is $\sqrt{u}\Big|_{\frac{1}{4}}^{1} = 1-\frac{1}{2} = \frac{1}{2}$.

Common pitfalls

- mixing a new variable with old limits
- changing limits through a non-one-to-one map without splitting
- losing a constant or minus sign in the differential

Verification

An antiderivative is $-\sqrt{1-x^2}$; upper minus lower gives $-\frac{1}{2}-(-1) = \frac{1}{2}$ again.

Method takeaway

A definite-integral substitution must transform the variable, differential, and endpoints as one coherent system; the substitution determines orientation.

Extension

Back-substitute before evaluating at the original endpoints as an independent verification route.

Source lineage: Open-text method adaptation · definite_integral_substitution · reference IDs: mit-18.01sc-definite-integral, openstax-calculus-v1-5.5. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | Fill in the blank | Synthesis | Classic-method variant

Q057 · Endpoint Substitution in an Exponential Fraction

Problem

Evaluate $\int_0^{\ln 2} \frac{e^x}{1+e^x}\,dx =$ ____.

Answer

$\ln \frac{3}{2}$

Knowledge points

- substitution in definite integrals
- transforming the variable, differential, and limits together
- orientation of transformed limits

Reading and method choice

Setting $u = 1 + e^x$ absorbs both the numerator and differential; the new endpoints are 2 and 3.

Detailed derivation

1. Set $u = 1 + e^x$, so $du = e^x\,dx$.
2. When $x = 0$, $u = 2$; when $x = \ln 2$, $u = 3$.
3. The integral becomes $\int_2^3 \frac{1}{u}\,du$.
4. Hence the value is $\ln 3 - \ln 2 = \ln \frac{3}{2}$.

Common pitfalls

- mixing a new variable with old limits
- changing limits through a non-one-to-one map without splitting
- losing a constant or minus sign in the differential

Verification

The integrand lies between $\frac{1}{2}$ and $\frac{2}{3}$ on an interval of length $\ln 2$; $\ln \frac{3}{2}$ lies between the corresponding bounds.

Method takeaway

A definite-integral substitution must transform the variable, differential, and endpoints as one coherent system; the substitution determines orientation.

Extension

Back-substitute before evaluating at the original endpoints as an independent verification route.

Source lineage: Classic-method variant · definite_integral_substitution · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | Single choice | Synthesis | Open-text method adaptation

Q058 · The Boundary Term in Integration by Parts

Problem

Let u, v be continuously differentiable on $[a, b]$. Which formula for definite integration by parts is correct?

A. $\int_a^b uv'\,dx = uv - \int_a^b u'v\,dx$

B. $\int_a^b uv'\,dx = [uv]_a^b + \int_a^b u'v\,dx$

C. $\int_a^b uv'\,dx = [uv]_a^b - \int_a^b u'v\,dx$

D. $\int_a^b uv'\,dx = \int_a^b u'v\,dx$

Answer

C

Knowledge points

- integration by parts for definite integrals
- boundary terms
- factor choice and reduction relations

Reading and method choice

Integrate the product differential $d(uv) = u\,dv + v\,du$ over $[a, b]$ and move $\int v\,du$ to the other side.

Detailed derivation

1. The product rule gives $(uv)' = u'v + uv'$.
2. Integrating over $[a, b]$ yields $[uv]_a^b = \int_a^b u'v\,dx + \int_a^b uv'\,dx$.
3. Rearrangement gives $\int_a^b uv'\,dx = [uv]_a^b - \int_a^b u'v\,dx$.
4. This is choice C; the others omit the boundary term or use an incorrect sign.

Common pitfalls

- omitting or mis-evaluating the boundary term
- copying the indefinite formula without endpoints
- failing to check that the new integral is simpler

Verification

With $u = x$, $v = x$, and $[a, b] = [0, 1]$, both sides of C equal $\frac{1}{2}$.

Method takeaway

Integration by parts trades the original integral for a controlled boundary term and a simpler definite integral.

Extension

Differentiate a parameterized closed form or estimate numerically to check the sign and scale of boundary terms.

Source lineage: Open-text method adaptation · definite_integral_by_parts · reference IDs: mit-18.01sc-definite-integral, openstax-calculus-v2-3.1. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard **Calculation** **Synthesis** **Classic-method variant**

Q059 · Integration by Parts with a Logarithmic Factor

Problem

Evaluate $\int_0^1 x \ln(1+x)\,dx$.

Answer

$\frac{1}{4}$

Knowledge points

- integration by parts for definite integrals
- boundary terms
- factor choice and reduction relations

Reading and method choice

Differentiate the logarithmic factor and integrate the algebraic factor; polynomial division simplifies the resulting $\frac{x^2}{1+x}$.

Detailed derivation

1. Choose $u = \ln(1+x)$ and $dv = x\,dx$, so $du = \frac{dx}{1+x}$ and $v = \frac{x^2}{2}$.
2. Integration by parts gives $\int_0^1 x \ln(1+x)\,dx = \frac{x^2}{2}\ln(1+x)\Big|_0^1 - \frac{1}{2}\int_0^1 \frac{x^2}{1+x}\,dx$.
3. The boundary term is $\frac{1}{2}\ln 2$, and $\frac{x^2}{1+x} = x - 1 + \frac{1}{1+x}$.
4. Hence $\int_0^1 \frac{x^2}{1+x}\,dx = \left(\frac{x^2}{2} - x + \ln(1+x)\right)\Big|_0^1 = \ln 2 - \frac{1}{2}$.
5. Substitution back gives $\frac{1}{2}\ln 2 - \frac{1}{2}\left(\ln 2 - \frac{1}{2}\right) = \frac{1}{4}$.

Common pitfalls

- omitting or mis-evaluating the boundary term
- copying the indefinite formula without endpoints
- failing to check that the new integral is simpler

Verification

Since $0 < \ln(1+x) < x$ for $0 < x \le 1$, one has $0 < I < \int_0^1 x^2\,dx = \frac{1}{3}$; the value $\frac{1}{4}$ has the correct sign and scale.

Method takeaway

Integration by parts trades the original integral for a controlled boundary term and a simpler definite integral.

Extension

Differentiate a parameterized closed form or estimate numerically to check the sign and scale of boundary terms.

Source lineage: Classic-method variant · definite_integral_by_parts · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Proof Synthesis Original synthesis / diagnosis

Q060 · A Chain-Rule Proof of Definite Substitution

Problem

Let f be continuous on an interval containing $\varphi([\alpha,\beta])$, and let φ be continuously differentiable on $[\alpha,\beta]$. Prove $\int_{\alpha}^{\beta} f(\varphi(t))\varphi'(t)\,dt = \int_{\varphi(\alpha)}^{\varphi(\beta)} f(u)\,du$, and explain why monotonicity of φ is unnecessary.

Answer

See the proof; the result follows directly from an antiderivative and the chain rule.

Knowledge points

- substitution in definite integrals
- transforming the variable, differential, and limits together
- orientation of transformed limits

Reading and method choice

Continuity gives an antiderivative of f, and the derivative of the composite $F \circ \varphi$ is exactly the left integrand, so Newton-Leibniz at the endpoints suffices.

Detailed derivation

1. Choose an antiderivative F of f, so $F'(u) = f(u)$.
2. The chain rule gives $\frac{d}{dt}F(\varphi(t)) = f(\varphi(t))\varphi'(t)$.
3. Newton-Leibniz makes the left side $F(\varphi(\beta)) - F(\varphi(\alpha))$.
4. Meanwhile, $\int_{\varphi(\alpha)}^{\varphi(\beta)} f(u)\,du = F(\varphi(\beta)) - F(\varphi(\alpha))$.
5. Only endpoint values and the chain rule were used; the identity remains valid if φ reverses direction internally, so monotonicity is unnecessary.

Common pitfalls

- mixing a new variable with old limits
- changing limits through a non-one-to-one map without splitting
- losing a constant or minus sign in the differential

Verification

With $f(u) = 1$, both sides equal $\varphi(\beta) - \varphi(\alpha)$, directly checking endpoint orientation and sign.

Method takeaway

A definite-integral substitution must transform the variable, differential, and endpoints as one coherent system; the substitution determines orientation.

Extension

Back-substitute before evaluating at the original endpoints as an independent verification route.

Source lineage: Original synthesis / diagnosis · definite_integral_substitution · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced | Multiple choice | Synthesis | Original synthesis / diagnosis

Q061 · Substitution Identities with a General Function

Problem

Let f be continuous on the relevant intervals and let $a > 0$. Which identities hold?

A. $\displaystyle\int_0^1 x f(x^2)\,dx = \frac{1}{2}\int_0^1 f(u)\,du$

B. $\displaystyle\int_{-1}^1 x f(x^2)\,dx = \int_0^1 f(u)\,du$

C. $\displaystyle\int_{-1}^1 f(x^2)\,dx = \int_0^1 \frac{f(u)}{\sqrt{u}}\,du$

D. $\displaystyle\int_0^a f\left(\frac{x}{a}\right)\,dx = a\int_0^1 f(u)\,du$

Answer

A, C, and D.

Knowledge points

- substitution in definite integrals
- transforming the variable, differential, and limits together
- orientation of transformed limits

Reading and method choice

A and B contrast $u = x^2$ on a half interval and a symmetric interval; C first uses evenness, and D is a linear scaling.

Detailed derivation

1. In A, $u = x^2$ and $du = 2x\,dx$ give $\frac{1}{2}\int_0^1 f(u)\,du$ directly.
2. In B, $x f(x^2)$ is odd, so the left side is 0 while the right side need not be 0; B is false.
3. In C, $f(x^2)$ is even; write $2\int_0^1 f(x^2)\,dx$ and then set $u = x^2$ to obtain $\int_0^1 \frac{f(u)}{\sqrt{u}}\,du$.
4. In D, $u = \frac{x}{a}$ gives $dx = a\,du$ and maps $0, a$ to $0, 1$, so D holds.

Common pitfalls

- mixing a new variable with old limits
- changing limits through a non-one-to-one map without splitting
- losing a constant or minus sign in the differential

Verification

For $f \equiv 1$, B gives 0 on the left and 1 on the right; A, C, and D give matching values $\frac{1}{2}$, 2, and a.

Method takeaway

A definite-integral substitution must transform the variable, differential, and endpoints as one coherent system; the substitution determines orientation.

Extension

Back-substitute before evaluating at the original endpoints as an independent verification route.

Source lineage: Original synthesis / diagnosis · definite_integral_substitution · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced **True/false with justification** **Synthesis** **Original synthesis / diagnosis**

Q062 · A Nonmonotone Map Can Still Satisfy the Substitution Formula

Problem

True or false: if f is continuous, then even though $u = x^2$ is not monotone on $[-1, 1]$, the definite-integral substitution formula still gives $\displaystyle\int_{-1}^{1} x f(x^2)\, dx = \frac{1}{2}\int_{1}^{1} f(u)\, du = 0$ directly.

Answer

True.

Knowledge points

- substitution in definite integrals
- transforming the variable, differential, and limits together
- orientation of transformed limits

Reading and method choice

The integrand contains exactly the factor $\varphi'(x) = 2x$. The definite-integral substitution theorem follows from an antiderivative and the chain rule, so monotonicity of $\varphi(x) = x^2$ is unnecessary.

Detailed derivation

1. Let F be an antiderivative of f. The chain rule gives $\frac{d}{dx}F(x^2) = 2xf(x^2)$.
2. Hence $\int_{-1}^{1} xf(x^2)\, dx = \frac{1}{2}\left[F(x^2)\right]_{-1}^{1}$.
3. Both endpoints give $x^2 = 1$, so the value is $\frac{1}{2}(F(1) - F(1)) = 0$.
4. This is exactly $\frac{1}{2}\int_{1}^{1} f(u)\, du = 0$, and no monotonicity of x^2 was used.
5. As a check, $f(x^2)$ is even, so $xf(x^2)$ is odd and its symmetric-interval integral is 0.

Common pitfalls

- mixing a new variable with old limits
- changing limits through a non-one-to-one map without splitting
- losing a constant or minus sign in the differential

Verification

For $f(u) = e^u$, take $F(u) = e^u$; the endpoint difference is $\frac{1}{2}(e - e) = 0$, matching the oddness check.

Method takeaway

A definite-integral substitution must transform the variable, differential, and endpoints as one coherent system; the substitution determines orientation.

Extension

Back-substitute before evaluating at the original endpoints as an independent verification route.

Source lineage: Original synthesis / diagnosis · definite_integral_substitution · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced | Calculation | Synthesis | Classic-method variant

Q063 · Rational Decomposition after Substitution

Problem

Evaluate $\displaystyle\int_0^1 \frac{x^3}{(1+x^2)^2}\,dx$.

Answer

$\frac{1}{2}\ln 2 - \frac{1}{4}$

Knowledge points

- substitution in definite integrals
- transforming the variable, differential, and limits together
- orientation of transformed limits

Reading and method choice

After setting $u = 1 + x^2$, rewrite x^2 as $u - 1$ so that no old variable remains.

Detailed derivation

1. Set $u = 1 + x^2$, so $du = 2x\,dx$ and $x^2 = u - 1$.
2. The endpoints $x = 0, 1$ become $u = 1, 2$.
3. The integral becomes $\frac{1}{2}\int_1^2 \frac{u-1}{u^2}\,du = \frac{1}{2}\int_1^2 \left(\frac{1}{u} - \frac{1}{u^2}\right) du$.
4. An antiderivative is $\frac{1}{2}\left(\ln u + \frac{1}{u}\right)$.
5. Endpoint evaluation gives $\frac{1}{2}\left(\ln 2 + \frac{1}{2} - 1\right) = \frac{1}{2}\ln 2 - \frac{1}{4}$.

Common pitfalls

- mixing a new variable with old limits
- changing limits through a non-one-to-one map without splitting
- losing a constant or minus sign in the differential

Verification

The answer is approximately $0.0966 > 0$, consistent with the nonnegative and relatively small integrand on $[0, 1]$.

Method takeaway

A definite-integral substitution must transform the variable, differential, and endpoints as one coherent system; the substitution determines orientation.

Extension

Back-substitute before evaluating at the original endpoints as an independent verification route.

Source lineage: Classic-method variant · definite_integral_substitution · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced **Calculation** **Synthesis** **Classic-method variant**

Q064 · A Linear Factor Times Cosine

Problem

Evaluate $\int_0^{\frac{\pi}{2}} x \cos x \, dx$.

Answer

$\frac{\pi}{2} - 1$

Knowledge points

- integration by parts for definite integrals
- boundary terms
- factor choice and reduction relations

Reading and method choice

Choose $u = x$ and $dv = \cos x \, dx$; the upper boundary contributes $\frac{\pi}{2}$, and the remaining sine integral contributes 1.

Detailed derivation

1. Let $u = x$ and $dv = \cos x \, dx$, so $du = dx$ and $v = \sin x$.
2. Integration by parts gives $\int_0^{\frac{\pi}{2}} x \cos x \, dx = x \sin x|_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \sin x \, dx$.
3. The boundary term is $\frac{\pi}{2}$.
4. The remaining integral is 1, so the value is $\frac{\pi}{2} - 1$.

Common pitfalls

- omitting or mis-evaluating the boundary term
- copying the indefinite formula without endpoints
- failing to check that the new integral is simpler

Verification

The integrand is nonnegative and $\frac{\pi}{2} - 1 \approx 0.571 > 0$; differentiating $x \sin x + \cos x$ also recovers $x \cos x$.

Method takeaway

Integration by parts trades the original integral for a controlled boundary term and a simpler definite integral.

Extension

Differentiate a parameterized closed form or estimate numerically to check the sign and scale of boundary terms.

Source lineage: Classic-method variant · definite_integral_by_parts · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Proof Synthesis Original synthesis / diagnosis

Q065 · A Weighted Integral with Midpoint Symmetry

Problem

Let $a > 0$ and let f be continuous on $[0, a]$ with $f(a - x) = f(x)$. Prove $\displaystyle\int_0^a x f(x)\, dx = \frac{a}{2}\int_0^a f(x)\, dx$.

Answer

See the proof.

Knowledge points

- substitution in definite integrals
- transforming the variable, differential, and limits together
- orientation of transformed limits

Reading and method choice

Call the weighted integral I and reflect with $u = a - x$; symmetry restores $f(u)$, after which adding the two representations removes the weight.

Detailed derivation

1. Let $I = \int_0^a x f(x)\, dx$.
2. Set $u = a - x$, so $dx = -du$ and endpoints $0, a$ become $a, 0$.
3. Using $f(a - u) = f(u)$ gives $I = \int_0^a (a - u) f(u)\, du$.
4. Add this to $I = \int_0^a u f(u)\, du$ to obtain $2I = a \int_0^a f(u)\, du$.
5. Division by 2 proves the formula.

Common pitfalls

- mixing a new variable with old limits
- changing limits through a non-one-to-one map without splitting
- losing a constant or minus sign in the differential

Verification

For $f \equiv 1$, the left side is $\frac{a^2}{2}$ and the right side is $\frac{a}{2} \cdot a = \frac{a^2}{2}$.

Method takeaway

A definite-integral substitution must transform the variable, differential, and endpoints as one coherent system; the substitution determines orientation.

Extension

Back-substitute before evaluating at the original endpoints as an independent verification route.

Source lineage: Original synthesis / diagnosis · definite_integral_substitution · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced | Parameter / synthesis / application | Synthesis | Original synthesis / diagnosis

Q066 · A Parameterized Semicircle-Radical Integral

Problem

Let $a > 0$. Evaluate $I(a) = \displaystyle\int_0^a x\sqrt{a^2 - x^2}\,dx$ and identify its scaling power in a.

Answer

$I(a) = \frac{a^3}{3}$, scaling cubically in a.

Knowledge points

- substitution in definite integrals
- transforming the variable, differential, and limits together
- orientation of transformed limits

Reading and method choice

The quadratic inside the radical pairs with $x\,dx$; the substitution $u = a^2 - x^2$ sends the endpoints from a^2 to 0.

Detailed derivation

1. Set $u = a^2 - x^2$, so $du = -2x\,dx$.
2. When $x = 0$, $u = a^2$; when $x = a$, $u = 0$.
3. Thus $I(a) = -\frac{1}{2}\int_{a^2}^{0} u^{\frac{1}{2}}\,du = \frac{1}{2}\int_0^{a^2} u^{\frac{1}{2}}\,du$.
4. Evaluation gives $I(a) = \frac{1}{2} \cdot \frac{2}{3}(a^2)^{\frac{3}{2}}$.
5. Since $a > 0$, $(a^2)^{\frac{3}{2}} = a^3$, so $I(a) = \frac{a^3}{3}$ and the scaling is cubic.

Common pitfalls

- mixing a new variable with old limits
- changing limits through a non-one-to-one map without splitting
- losing a constant or minus sign in the differential

Verification

Alternatively, $x = at$ extracts $a \cdot a \cdot a = a^3$ and gives $I(a) = a^3 \int_0^1 t\sqrt{1-t^2}\,dt = \frac{a^3}{3}$.

Method takeaway

A definite-integral substitution must transform the variable, differential, and endpoints as one coherent system; the substitution determines orientation.

Extension

Back-substitute before evaluating at the original endpoints as an independent verification route.

Source lineage: Original synthesis / diagnosis · definite_integral_substitution · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced | Error diagnosis | Synthesis | Original synthesis / diagnosis

Q067 · Diagnosing a Non-injective Substitution

Problem

A student evaluates $J = \displaystyle\int_{-1}^{1} |x|e^{x^2}\,dx$ by setting $u = x^2$ on the whole interval and writes $J = \frac{1}{2}\int_1^1 e^u\,du = 0$ because both transformed endpoints are 1. Identify the error and find the correct value.

Answer

$J = e - 1$; the error is treating $|x|\,dx$ as $\frac{1}{2}du$ on both branches and ignoring that $|x| = -x$ on the negative half-axis.

Knowledge points

- substitution in definite integrals
- transforming the variable, differential, and limits together
- orientation of transformed limits

Reading and method choice

The two branches of x^2 have opposite orientations, while the absolute value changes the sign relation in the differential. Use evenness first or split at 0.

Detailed derivation

1. $|x|e^{x^2}$ is even, so $J = 2\int_0^1 xe^{x^2}\,dx$.
2. On $[0,1]$, set $u = x^2$. Then $du = 2x\,dx$, and endpoints $0, 1$ remain $0, 1$.
3. Therefore $J = 2 \cdot \frac{1}{2}\int_0^1 e^u\,du = e^u\big|_0^1 = e - 1$.
4. The student's global step omits $|x| = -x$ on the negative half-axis, where $|x|\,dx = -\frac{1}{2}du$.
5. Since the original integrand is nonnegative and not identically zero, the claimed value 0 also fails an immediate sign check.

Common pitfalls

- mixing a new variable with old limits
- changing limits through a non-one-to-one map without splitting
- losing a constant or minus sign in the differential

Verification

Splitting verifies that each half contributes $\frac{1}{2}(e-1)$, so the total is $e-1>0$.

Method takeaway

A definite-integral substitution must transform the variable, differential, and endpoints as one coherent system; the substitution determines orientation.

Extension

Back-substitute before evaluating at the original endpoints as an independent verification route.

Source lineage: Original synthesis / diagnosis · definite_integral_substitution · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Calculation Synthesis Classic-method variant

Q068 · A Definite Integral of Arctangent

Problem

Evaluate $\int_0^1 \arctan x\,dx$.

Answer

$\frac{\pi}{4} - \frac{1}{2}\ln 2$

Knowledge points

- integration by parts for definite integrals
- boundary terms
- factor choice and reduction relations

Reading and method choice

Use 1 as the easily integrated factor and take $u = \arctan x$; integration by parts produces $\frac{x}{1+x^2}$, which is logarithmic.

Detailed derivation

1. Choose $u = \arctan x$ and $dv = dx$, so $du = \frac{dx}{1+x^2}$ and $v = x$.
2. Integration by parts gives $\int_0^1 \arctan x\,dx = \left. x\arctan x\right|_0^1 - \int_0^1 \frac{x}{1+x^2}\,dx$.
3. The boundary term is $\frac{\pi}{4}$.
4. With $w = 1 + x^2$, the remaining integral is $\frac{1}{2}\ln 2$.
5. Hence the value is $\frac{\pi}{4} - \frac{1}{2}\ln 2$.

Common pitfalls

- omitting or mis-evaluating the boundary term
- copying the indefinite formula without endpoints
- failing to check that the new integral is simpler

Verification

Since $0 \le \arctan x \le \frac{\pi}{4}$, the integral lies between 0 and $\frac{\pi}{4}$; the value is approximately 0.439.

Method takeaway

Integration by parts trades the original integral for a controlled boundary term and a simpler definite integral.

Extension

Differentiate a parameterized closed form or estimate numerically to check the sign and scale of boundary terms.

Source lineage: Classic-method variant · definite_integral_by_parts · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced **Proof** **Synthesis** **Classic-method variant**

Q069 · The Wallis Integral Recurrence

Problem

For an integer $n \geq 2$, let $I_n = \displaystyle\int_0^{\frac{\pi}{2}} \sin^n x\,dx$. Prove $I_n = \dfrac{n-1}{n} I_{n-2}$ and state I_0, I_1.

Answer

$I_n = \frac{n-1}{n} I_{n-2}$, $I_0 = \frac{\pi}{2}$, **and** $I_1 = 1$.

Knowledge points

- integration by parts for definite integrals
- boundary terms
- factor choice and reduction relations

Reading and method choice

Use one factor $\sin x$ as dv and differentiate $\sin^{n-1} x$; after $\cos^2 x = 1 - \sin^2 x$, the integral I_n reappears.

Detailed derivation

1. Write $I_n = \int_0^{\frac{\pi}{2}} \sin^{n-1} x \sin x\,dx$.
2. Take $u = \sin^{n-1} x$ and $dv = \sin x\,dx$, giving $du = (n-1)\sin^{n-2} x \cos x\,dx$ and $v = -\cos x$.
3. The boundary term $[-\sin^{n-1} x \cos x]_0^{\frac{\pi}{2}}$ is 0.
4. Therefore $I_n = (n-1)\int_0^{\frac{\pi}{2}} \sin^{n-2} x \cos^2 x\,dx$.
5. Using $\cos^2 x = 1 - \sin^2 x$ gives $I_n = (n-1)(I_{n-2} - I_n)$.
6. Thus $nI_n = (n-1)I_{n-2}$; direct integration gives $I_0 = \frac{\pi}{2}$ and $I_1 = 1$.

Common pitfalls

- omitting or mis-evaluating the boundary term
- copying the indefinite formula without endpoints
- failing to check that the new integral is simpler

Verification

The recurrence gives $I_2 = \frac{1}{2}I_0 = \frac{\pi}{4}$, agreeing with direct integration using the power-reduction identity.

Method takeaway

Integration by parts trades the original integral for a controlled boundary term and a simpler definite integral.

Extension

Differentiate a parameterized closed form or estimate numerically to check the sign and scale of boundary terms.

Source lineage: Classic-method variant · definite_integral_by_parts · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Proof Synthesis Original synthesis / diagnosis

Q070 · A Weighted-Derivative Identity from Integration by Parts

Problem

Let $a > 0$ and let f be continuously differentiable on $[0, a]$. Prove $\int_0^a x f'(x)\,dx = af(a) - \int_0^a f(x)\,dx$.

Answer

See the proof.

Knowledge points

- integration by parts for definite integrals
- boundary terms
- factor choice and reduction relations

Reading and method choice

Use x as the factor simplified by differentiation and $f'(x)\,dx$ as the directly integrable factor; the lower boundary vanishes because it is multiplied by x.

Detailed derivation

1. Choose $u = x$ and $dv = f'(x)\,dx$, so $du = dx$ and $v = f(x)$.
2. Definite integration by parts yields $\int_0^a x f'(x)\,dx = x f(x)|_0^a - \int_0^a f(x)\,dx$.
3. The upper boundary is $af(a)$.
4. The lower boundary is $0 \cdot f(0) = 0$.
5. Substitution gives the desired identity.

Common pitfalls

- omitting or mis-evaluating the boundary term
- copying the indefinite formula without endpoints
- failing to check that the new integral is simpler

Verification

For $f(x) = x^2$, the left side is $\int_0^a 2x^2 dx = \frac{2a^3}{3}$ and the right side is $a^3 - \frac{a^3}{3} = \frac{2a^3}{3}$.

Method takeaway

Integration by parts trades the original integral for a controlled boundary term and a simpler definite integral.

Extension

Differentiate a parameterized closed form or estimate numerically to check the sign and scale of boundary terms.

Source lineage: Original synthesis / diagnosis · definite_integral_by_parts · reference IDs: mit-18.01sc-definite-integral. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced | True/false with justification | Synthesis | Open-text method adaptation

Q071 · The Power Test on an Infinite Interval

Problem

For real p, determine whether the following statement is true: the improper integral $\displaystyle\int_1^{+\infty} \frac{dx}{x^p}$ converges if and only if $p > 1$.

Answer

True.

Knowledge points

- definition on an infinite interval
- truncation limits
- convergence values and tails

Reading and method choice

Handle $p = 1$ and $p \neq 1$ separately; convergence depends on whether the truncated antiderivative has a finite limit as $R \to +\infty$.

Detailed derivation

1. By definition, examine $\lim_{R\to+\infty} \int_1^R x^{-p}\, dx$.
2. For $p \neq 1$, the truncated integral is $\frac{R^{1-p}-1}{1-p}$.
3. If $p > 1$, then $R^{1-p} \to 0$ and the limit is $\frac{1}{p-1}$; if $p < 1$, it diverges to $+\infty$.
4. For $p = 1$, the truncated integral is $\ln R \to +\infty$.
5. Thus convergence occurs exactly when $p > 1$.

Common pitfalls

- substituting infinity as if it were an endpoint
- failing to split a two-sided infinite interval into independent limits
- confusing a symmetric principal value with ordinary convergence

Verification

For $p = 2$ the value is 1, while the boundary case $p = 1$ gives the divergent $\ln R$, checking both the range and its threshold.

Method takeaway

An infinite endpoint must be truncated before taking a limit, and a two-sided integral requires both one-sided pieces to converge independently.

Extension

Vary the truncation scheme to expose the difference between a principal value and an ordinary improper integral.

Source lineage: Open-text method adaptation · improper_integral_infinite_interval · reference IDs: mit-18.01sc-improper-integrals, openstax-calculus-v2-3.7. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced **Calculation** **Synthesis** **Open-text method adaptation**

Q072 · A Basic Improper Integral on an Infinite Interval

Problem

Evaluate $\int_1^{+\infty} \frac{dx}{x^2}$.

Answer

1

Knowledge points

- definition on an infinite interval
- truncation limits
- convergence values and tails

Reading and method choice

Replace $+\infty$ by a finite upper limit R, evaluate, and then let $R \to +\infty$; infinity is not an endpoint value to substitute directly.

Detailed derivation

1. By definition, write $\lim_{R\to+\infty} \int_1^R x^{-2}\,dx$.
2. An antiderivative is $-x^{-1}$.
3. The truncated integral is $-\frac{1}{x}\Big|_1^R = 1 - \frac{1}{R}$.
4. As $R \to +\infty$, the limit is 1, so the improper integral converges to 1.

Common pitfalls

- substituting infinity as if it were an endpoint
- failing to split a two-sided infinite interval into independent limits
- confusing a symmetric principal value with ordinary convergence

Verification

The tail $\int_A^{+\infty} x^{-2}dx = \frac{1}{A} \to 0$, independently matching the vanishing-tail property of a convergent improper integral.

Method takeaway

An infinite endpoint must be truncated before taking a limit, and a two-sided integral requires both one-sided pieces to converge independently.

Extension

Vary the truncation scheme to expose the difference between a principal value and an ordinary improper integral.

Source lineage: Open-text method adaptation · improper_integral_infinite_interval · reference IDs: mit-18.01sc-improper-integrals, openstax-calculus-v2-3.7. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Calculation Synthesis Open-text method adaptation

Q073 · An Integrable Power Singularity at an Endpoint

Problem

Evaluate $\int_0^1 \frac{dx}{\sqrt{x}}$.

Answer

2

Knowledge points

- endpoint-singular improper integrals
- one-sided limits
- power and logarithmic endpoint behavior

Reading and method choice

The integrand is unbounded at $x = 0$, so truncate from the right; the exponent $-\frac{1}{2} > -1$ indicates integrability.

Detailed derivation

1. By definition, write $\lim_{\varepsilon\to 0^+} \int_\varepsilon^1 x^{-\frac{1}{2}}\, dx$.
2. An antiderivative is $2\sqrt{x}$.
3. The truncated integral equals $2 - 2\sqrt{\varepsilon}$.
4. As $\varepsilon \to 0^+$, $\sqrt{\varepsilon} \to 0$.
5. Therefore the improper integral converges to 2.

Common pitfalls

- substituting the singular endpoint directly
- ignoring the direction of the limit
- judging from an antiderivative without evaluating the truncation limit

Verification

Although the integrand is unbounded, the truncated area $2-2\sqrt{\varepsilon}$ has a finite limit, showing that unbounded does not mean nonintegrable.

Method takeaway

Both the value and convergence of an endpoint-singular integral are determined by its one-sided truncation limit.

Extension

Comparison with x^{-p} or $|\ln x|$ often reveals local integrability near the endpoint.

Source lineage: Open-text method adaptation · improper_integral_singular_endpoint · reference IDs: mit-18.01sc-improper-integrals, openstax-calculus-v2-3.7. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Proof Synthesis Classic-method variant

Q074 · Proving Logarithmic Divergence by Comparison

Problem

Prove that $\int_1^{+\infty} \frac{dx}{x+1}$ diverges.

Answer

It diverges to $+\infty$.

Knowledge points

- comparison tests for improper integrals
- limit comparison
- absolute and conditional convergence

Reading and method choice

For $x \geq 1$, bound below by a constant multiple of the divergent benchmark $\frac{1}{x}$; a nonnegative integral above a divergent one also diverges.

Detailed derivation

1. For $x \geq 1$, $x + 1 \leq 2x$.
2. Taking reciprocals of positive quantities gives $\frac{1}{x+1} \geq \frac{1}{2x}$.
3. For every $R > 1$, $\int_1^R \frac{dx}{x+1} \geq \frac{1}{2} \int_1^R \frac{dx}{x} = \frac{1}{2} \ln R$.
4. As $R \to +\infty$, $\frac{1}{2} \ln R \to +\infty$.
5. The comparison test therefore proves divergence to $+\infty$.

Common pitfalls

- reversing the comparison inequality
- using the finite-positive limit conclusion when the ratio tends to zero or infinity
- proving convergence without checking absolute convergence

Verification

Direct truncation gives $\ln(R+1) - \ln 2 \to +\infty$, agreeing with the comparison argument.

Method takeaway

A convergence comparison must state nonnegativity, the tail or neighborhood being compared, and the benchmark integral.

Extension

Use the same benchmark to build both upper and lower comparisons and clarify the direction of sufficient and necessary conditions.

Source lineage: Classic-method variant · improper_integral_convergence_tests · reference IDs: mit-18.01sc-improper-integrals. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced | Parameter / synthesis / application | Synthesis | Original synthesis / diagnosis

Q075 · A Principal Value Exists but the Ordinary Integral Diverges

Problem

Discuss the ordinary improper integral and the Cauchy principal value of $\displaystyle\int_{-1}^{1} \frac{dx}{x}$.

Answer

The ordinary improper integral diverges, while PV $\displaystyle\int_{-1}^{1} \frac{dx}{x} = 0$.

Knowledge points

- interior singularities
- independent convergence on both sides
- Cauchy principal value versus ordinary convergence

Reading and method choice

The interior singularity at 0 splits the interval into two independent improper integrals; only the principal value uses one symmetric truncation parameter.

Detailed derivation

1. The left part is $\lim_{c\to 0^-} \int_{-1}^{c} \frac{dx}{x} = \lim_{c\to 0^-} \ln|c| = -\infty$.
2. The right part is $\lim_{d\to 0^+} \int_{d}^{1} \frac{dx}{x} = \lim_{d\to 0^+}(-\ln d) = +\infty$.
3. Ordinary convergence requires both sides to have finite limits, so the integral diverges; $-\infty$ and $+\infty$ cannot be canceled.
4. The principal value is $\lim_{\varepsilon\to 0^+} \left(\int_{-1}^{-\varepsilon} \frac{dx}{x} + \int_{\varepsilon}^{1} \frac{dx}{x}\right)$.
5. The two terms cancel exactly for every ε, hence the principal value is 0.

Common pitfalls

- applying Newton-Leibniz across a singularity
- canceling positive and negative infinities
- reporting a symmetric truncation limit as the ordinary integral

Verification

Using unequal cutoffs, such as distances ε on the left and 2ε on the right, makes the sum tend to $-\ln 2$ rather than 0, showing the lack of an ordinary limit.

Method takeaway

An interior singularity splits the interval; divergence on either side makes the ordinary improper integral divergent.

Extension

If a symmetric principal value exists, label it explicitly as PV rather than omitting the qualifier.

Source lineage: Original synthesis / diagnosis · improper_integral_interior_singularity · reference IDs: mit-18.01sc-improper-integrals. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced | Calculation | Synthesis | Classic-method variant

Q076 · A Cube-Root Endpoint Singularity

Problem

Evaluate $\int_0^1 \frac{dx}{\sqrt[3]{x}}$.

Answer

$\frac{3}{2}$

Knowledge points

- endpoint-singular improper integrals
- one-sided limits
- power and logarithmic endpoint behavior

Reading and method choice

Write the integrand as $x^{-\frac{1}{3}}$ and truncate to the right of 0; the antiderivative has exponent $\frac{2}{3} > 0$.

Detailed derivation

1. By definition, use $\lim_{\varepsilon\to 0^+} \int_\varepsilon^1 x^{-\frac{1}{3}}\, dx$.
2. An antiderivative is $\frac{3}{2} x^{\frac{2}{3}}$.
3. The truncated integral is $\frac{3}{2}\left(1 - \varepsilon^{\frac{2}{3}}\right)$.
4. Since $\varepsilon^{\frac{2}{3}} \to 0$, the limit is $\frac{3}{2}$.

Common pitfalls

- substituting the singular endpoint directly
- ignoring the direction of the limit
- judging from an antiderivative without evaluating the truncation limit

Verification

It agrees with the endpoint power test: $\int_0^1 x^{-p} dx$ converges for $p < 1$, here $p = \frac{1}{3}$.

Method takeaway

Both the value and convergence of an endpoint-singular integral are determined by its one-sided truncation limit.

Extension

Comparison with x^{-p} or $|\ln x|$ often reveals local integrability near the endpoint.

Source lineage: Classic-method variant · improper_integral_singular_endpoint · reference IDs: mit-18.01sc-improper-integrals. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Calculation Synthesis Classic-method variant

Q077 · The Parameter Sign in Exponential Decay

Problem

For real a, determine convergence and evaluate $I(a) = \int_0^{+\infty} e^{-ax}\,dx$.

Answer

It converges exactly for $a > 0$, with $I(a) = \frac{1}{a}$; it diverges for $a \le 0$.

Knowledge points

- definition on an infinite interval
- truncation limits
- convergence values and tails

Reading and method choice

The parameter a determines whether the exponential decays, stays constant, or grows, so cases must be separated before applying a formula.

Detailed derivation

1. If $a > 0$, the truncated integral is $\int_0^R e^{-ax} dx = \frac{1-e^{-aR}}{a}$.
2. Because $e^{-aR} \to 0$, $I(a) = \frac{1}{a}$.
3. If $a = 0$, the integrand is 1 and the truncated integral is $R \to +\infty$.
4. If $a < 0$, write $a = -c$ with $c > 0$; then the integrand is e^{cx} and its truncated integral grows exponentially.
5. Thus convergence occurs only for $a > 0$.

Common pitfalls

- substituting infinity as if it were an endpoint
- failing to split a two-sided infinite interval into independent limits
- confusing a symmetric principal value with ordinary convergence

Verification

Within the convergence range, $u = ax$ gives $I(a) = \frac{1}{a}\int_0^{+\infty} e^{-u}du = \frac{1}{a}$, independently checking the scale factor.

Method takeaway

An infinite endpoint must be truncated before taking a limit, and a two-sided integral requires both one-sided pieces to converge independently.

Extension

Vary the truncation scheme to expose the difference between a principal value and an ordinary improper integral.

Source lineage: Classic-method variant · improper_integral_infinite_interval · reference IDs: mit-18.01sc-improper-integrals. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced | Calculation | Synthesis | Classic-method variant

Q078 · Cancellation of Logarithms at Infinity

Problem

Evaluate $\int_1^{+\infty} \frac{dx}{x(x+1)}$.

Answer

$\ln 2$

Knowledge points

- definition on an infinite interval
- truncation limits
- convergence values and tails

Reading and method choice

First decompose $\frac{1}{x(x+1)} = \frac{1}{x} - \frac{1}{x+1}$; the logarithms must be combined before taking the limit.

Detailed derivation

1. For finite $R > 1$, write $\int_1^R \left(\frac{1}{x} - \frac{1}{x+1}\right) dx$.
2. The truncated integral is $\ln \frac{x}{x+1}\Big|_1^R$.
3. This equals $\ln \frac{R}{R+1} - \ln \frac{1}{2}$.
4. As $R \to +\infty$, $\frac{R}{R+1} \to 1$, so the first term tends to 0.
5. Therefore the integral converges to $\ln 2$.

Common pitfalls

- substituting infinity as if it were an endpoint
- failing to split a two-sided infinite interval into independent limits
- confusing a symmetric principal value with ordinary convergence

Verification

Since $0 < \frac{1}{x(x+1)} < \frac{1}{x^2}$ for $x \geq 1$, the integral must converge and be below 1; indeed $0 < \ln 2 < 1$.

Method takeaway

An infinite endpoint must be truncated before taking a limit, and a two-sided integral requires both one-sided pieces to converge independently.

Extension

Vary the truncation scheme to expose the difference between a principal value and an ordinary improper integral.

Source lineage: Classic-method variant · improper_integral_infinite_interval · reference IDs: mit-18.01sc-improper-integrals. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Proof Synthesis Original synthesis / diagnosis

Q079 · The Cauchy Tail Criterion for an Improper Integral

Problem

Let f be continuous on every finite interval $[a, R]$. Prove that $\displaystyle\int_a^{+\infty} f(x)\,dx$ converges if and only if, for every $\varepsilon > 0$, there exists $M > a$ such that all $B > A \geq M$ satisfy $\left|\displaystyle\int_A^B f(x)\,dx\right| < \varepsilon$.

Answer

See the proof.

Knowledge points

- comparison tests for improper integrals
- limit comparison
- absolute and conditional convergence

Reading and method choice

Let $F(R) = \int_a^R f$; the stated tail condition is exactly the Cauchy condition for the real-valued function $F(R)$ as $R \to +\infty$.

Detailed derivation

1. Define $F(R) = \int_a^R f(x)\,dx$; for $B > A$, $F(B) - F(A) = \int_A^B f(x)\,dx$.
2. If the improper integral converges to L, choose M so that $|F(R) - L| < \frac{\varepsilon}{2}$ whenever $R \geq M$.
3. Then $|F(B) - F(A)| \leq |F(B) - L| + |F(A) - L| < \varepsilon$, proving the tail condition.
4. Conversely, the tail condition says that $F(R)$ satisfies the Cauchy condition as $R \to +\infty$.
5. Completeness of the real numbers gives a finite limit $L = \lim_{R\to+\infty} F(R)$.
6. By definition, this is convergence of $\int_a^{+\infty} f(x)\,dx$.

Common pitfalls

- reversing the comparison inequality
- using the finite-positive limit conclusion when the ratio tends to zero or infinity
- proving convergence without checking absolute convergence

Verification

For $f(x) = x^{-2}$, the tail has magnitude $\frac{1}{A} - \frac{1}{B} < \frac{1}{A}$; choosing $M > \frac{1}{\varepsilon}$ verifies the criterion.

Method takeaway

A convergence comparison must state nonnegativity, the tail or neighborhood being compared, and the benchmark integral.

Extension

Use the same benchmark to build both upper and lower comparisons and clarify the direction of sufficient and necessary conditions.

Source lineage: Original synthesis / diagnosis · improper_integral_convergence_tests · reference IDs: mit-18.01sc-improper-integrals. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced | Parameter / synthesis / application | Synthesis | Original synthesis / diagnosis

Q080 · Independent Splitting of a Two-sided Infinite Integral

Problem

Using the ordinary improper-integral definition, evaluate $\displaystyle\int_{-\infty}^{+\infty} \frac{dx}{1+x^2}$.

Answer

π

Knowledge points

- definition on an infinite interval
- truncation limits
- convergence values and tails

Reading and method choice

An ordinary two-sided integral must be split at a finite point, here 0, into two independent limits; both pieces converge.

Detailed derivation

1. By definition, split it as $\int_{-\infty}^{0} \frac{dx}{1+x^2} + \int_{0}^{+\infty} \frac{dx}{1+x^2}$.
2. The left part is $\lim_{A\to-\infty}$ arctan $x\big|_A^0 = 0 - \left(-\frac{\pi}{2}\right) = \frac{\pi}{2}$.
3. The right part is $\lim_{B\to+\infty}$ arctan $x\big|_0^B = \frac{\pi}{2} - 0 = \frac{\pi}{2}$.
4. Both pieces have finite limits, so the ordinary improper integral converges.
5. Their sum is π.

Common pitfalls

- substituting infinity as if it were an endpoint
- failing to split a two-sided infinite interval into independent limits
- confusing a symmetric principal value with ordinary convergence

Verification

The integrand is even, so the value is $2\int_0^{+\infty} \frac{dx}{1+x^2} = 2 \cdot \frac{\pi}{2} = \pi$, consistent with the independent split.

Method takeaway

An infinite endpoint must be truncated before taking a limit, and a two-sided integral requires both one-sided pieces to converge independently.

Extension

Vary the truncation scheme to expose the difference between a principal value and an ordinary improper integral.

Source lineage: Original synthesis / diagnosis · improper_integral_infinite_interval · reference IDs: mit-18.01sc-improper-integrals. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced **Error diagnosis** **Synthesis** **Original synthesis / diagnosis**

Q081 · Misusing Symmetry as Ordinary Convergence

Problem

A student argues that $\frac{x}{1+x^2}$ is odd and therefore $\int_{-\infty}^{+\infty} \frac{x}{1+x^2}\, dx = 0$. Diagnose the error and determine the ordinary integral and symmetric principal value.

Answer

The ordinary improper integral diverges; PV $\int_{-\infty}^{+\infty} \frac{x}{1+x^2}\, dx = 0$.

Knowledge points

- definition on an infinite interval
- truncation limits
- convergence values and tails

Reading and method choice

Oddness controls symmetric truncations but does not guarantee finite independent limits on the two infinite tails.

Detailed derivation

1. The right truncated integral is $\int_0^R \frac{x}{1+x^2} dx = \frac{1}{2} \ln(1+R^2) \to +\infty$.
2. The left truncated integral is $\int_{-R}^0 \frac{x}{1+x^2} dx = -\frac{1}{2} \ln(1+R^2) \to -\infty$.
3. Ordinary convergence requires both pieces to have finite limits, so the integral diverges.
4. The symmetric principal value is $\lim_{R\to+\infty} \int_{-R}^R \frac{x}{1+x^2} dx$.
5. Every symmetric truncation is 0 by oddness, so the principal value is 0.

Common pitfalls

- substituting infinity as if it were an endpoint
- failing to split a two-sided infinite interval into independent limits
- confusing a symmetric principal value with ordinary convergence

Verification

If the left and right cutoffs are R and $2R$, the sum tends to $\ln 2$ rather than 0, proving dependence on the truncation scheme.

Method takeaway

An infinite endpoint must be truncated before taking a limit, and a two-sided integral requires both one-sided pieces to converge independently.

Extension

Vary the truncation scheme to expose the difference between a principal value and an ordinary improper integral.

Source lineage: Original synthesis / diagnosis · improper_integral_infinite_interval · reference IDs: mit-18.01sc-improper-integrals. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Calculation Synthesis Classic-method variant

Q082 · The Integrable Logarithmic Singularity

Problem

Evaluate $\displaystyle\int_0^1 \ln x\, dx$.

Answer

-1

Knowledge points

- endpoint-singular improper integrals
- one-sided limits
- power and logarithmic endpoint behavior

Reading and method choice

$\ln x$ tends to $-\infty$ at 0, but the product $x \ln x$ in its antiderivative has a finite right limit.

Detailed derivation

1. By definition, write $\lim_{\varepsilon\to 0^+} \int_\varepsilon^1 \ln x\, dx$.
2. Integration by parts gives the antiderivative $x \ln x - x$.
3. The truncated integral is $-1 - \varepsilon \ln \varepsilon + \varepsilon$.
4. Use $\lim_{\varepsilon\to 0^+} \varepsilon \ln \varepsilon = 0$ and $\varepsilon \to 0$.
5. Thus the improper integral converges to -1.

Common pitfalls

- substituting the singular endpoint directly
- ignoring the direction of the limit
- judging from an antiderivative without evaluating the truncation limit

Verification

The substitution $x = e^{-t}$ gives $-\int_0^{+\infty} te^{-t} dt = -1$, providing an independent check.

Method takeaway

Both the value and convergence of an endpoint-singular integral are determined by its one-sided truncation limit.

Extension

Comparison with x^{-p} or $|\ln x|$ often reveals local integrability near the endpoint.

Source lineage: Classic-method variant · improper_integral_singular_endpoint · reference IDs: mit-18.01sc-improper-integrals. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard | Calculation | Synthesis | Classic-method variant

Q083 · A Substitution with Both Zero and Infinite Endpoints

Problem

Evaluate $\displaystyle\int_0^{+\infty} \frac{dx}{(1+x)\sqrt{x}}$.

Answer

π

Knowledge points

- definition on an infinite interval
- truncation limits
- convergence values and tails

Reading and method choice

The integral has an integrable square-root singularity at 0 and an infinite interval; $x = t^2$ removes the radical while preserving both improper endpoints.

Detailed derivation

1. For $0 < \varepsilon < R$, first consider $\int_\varepsilon^R \frac{dx}{(1+x)\sqrt{x}}$.
2. Set $x = t^2$ with $t > 0$, so $dx = 2t\,dt$ and $\sqrt{x} = t$.
3. The truncated integral becomes $2\int_{\sqrt{\varepsilon}}^{\sqrt{R}} \frac{dt}{1+t^2}$.
4. Its value is $2\left(\arctan\sqrt{R} - \arctan\sqrt{\varepsilon}\right)$.
5. Letting $R \to +\infty$ and $\varepsilon \to 0^+$ gives $2\left(\frac{\pi}{2} - 0\right) = \pi$.

Common pitfalls

- substituting infinity as if it were an endpoint
- failing to split a two-sided infinite interval into independent limits
- confusing a symmetric principal value with ordinary convergence

Verification

Near 0 the integrand is comparable to $x^{-\frac{1}{2}}$, and at infinity to $x^{-\frac{3}{2}}$; both endpoints meet the power-integral convergence conditions.

Method takeaway

An infinite endpoint must be truncated before taking a limit, and a two-sided integral requires both one-sided pieces to converge independently.

Extension

Vary the truncation scheme to expose the difference between a principal value and an ordinary improper integral.

Source lineage: Classic-method variant · improper_integral_infinite_interval · reference IDs: mit-18.01sc-improper-integrals. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard Proof Synthesis Classic-method variant

Q084 · The Comparison Test for Nonnegative Functions

Problem

Assume $0 \le f(x) \le g(x)$ for all $x \ge a$, with both functions integrable on finite intervals. Prove: if $\displaystyle\int_a^{+\infty} g(x)\,dx$ converges, then $\displaystyle\int_a^{+\infty} f(x)\,dx$ converges; if $\displaystyle\int_a^{+\infty} f(x)\,dx$ diverges, then $\displaystyle\int_a^{+\infty} g(x)\,dx$ diverges.

Answer

See the proof.

Knowledge points

- comparison tests for improper integrals
- limit comparison
- absolute and conditional convergence

Reading and method choice

Nonnegativity makes truncated integrals monotone in the upper limit; the upper comparison supplies boundedness, and the divergence statement is the contrapositive.

Detailed derivation

1. Define $F(R) = \int_a^R f(x)\,dx$ and $G(R) = \int_a^R g(x)\,dx$.
2. From $0 \le f \le g$, we have $0 \le F(R) \le G(R)$ for every $R > a$.
3. If $\int_a^{+\infty} g$ converges, then $G(R)$ is bounded; hence the increasing function $F(R)$ is also bounded.
4. Monotone bounded convergence gives a finite limit for $F(R)$, so $\int_a^{+\infty} f$ converges.
5. The second claim is the contrapositive: convergence of the larger integral would force convergence of the smaller, so divergence of the smaller forces divergence of the larger.

Common pitfalls

- reversing the comparison inequality
- using the finite-positive limit conclusion when the ratio tends to zero or infinity
- proving convergence without checking absolute convergence

Verification

For example, $f(x) = \frac{1}{x^2+x}$ and $g(x) = \frac{1}{x^2}$ satisfy $0 < f < g$, and convergence of the larger integral forces convergence of the smaller.

Method takeaway

A convergence comparison must state nonnegativity, the tail or neighborhood being compared, and the benchmark integral.

Extension

Use the same benchmark to build both upper and lower comparisons and clarify the direction of sufficient and necessary conditions.

Source lineage: Classic-method variant · improper_integral_convergence_tests · reference IDs: mit-18.01sc-improper-integrals. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard | Parameter / synthesis / application | Synthesis | Original synthesis / diagnosis

Q085 · An Infinite Integral with a Parameterized Positive Quadratic

Problem

Let $a > 0$. Evaluate $I(a) = \displaystyle\int_0^{+\infty} \frac{dx}{x^2 + 2ax + 2a^2}$.

Answer

$I(a) = \frac{\pi}{4a}$

Knowledge points

- definition on an infinite interval
- truncation limits
- convergence values and tails

Reading and method choice

Complete the square as $(x + a)^2 + a^2$; $a > 0$ fixes the orientation and the positive arctangent scale factor.

Detailed derivation

1. Complete the square: $x^2 + 2ax + 2a^2 = (x + a)^2 + a^2$.
2. Set $u = \frac{x+a}{a}$, so $dx = a\,du$.
3. Because $a > 0$, $x = 0$ gives $u = 1$ and $x \to +\infty$ gives $u \to +\infty$.
4. The integral becomes $\frac{1}{a} \int_1^{+\infty} \frac{du}{1+u^2}$.
5. Thus $I(a) = \frac{1}{a}\left(\frac{\pi}{2} - \frac{\pi}{4}\right) = \frac{\pi}{4a}$.

Common pitfalls

- substituting infinity as if it were an endpoint
- failing to split a two-sided infinite interval into independent limits
- confusing a symmetric principal value with ordinary convergence

Verification

The direct scaling $x = at$ shows that the integral must scale as a^{-1}; $\frac{\pi}{4a}$ has exactly this dependence.

Method takeaway

An infinite endpoint must be truncated before taking a limit, and a two-sided integral requires both one-sided pieces to converge independently.

Extension

Vary the truncation scheme to expose the difference between a principal value and an ordinary improper integral.

Source lineage: Original synthesis / diagnosis · improper_integral_infinite_interval · reference IDs: mit-18.01sc-improper-integrals. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard | Error diagnosis | Synthesis | Original synthesis / diagnosis

Q086 · Applying Newton-Leibniz across a Second-order Singularity

Problem

A student writes $\int_0^2 \frac{dx}{(x-1)^2} = -\frac{1}{x-1}\Big|_0^2 = -2$. Identify all errors and give the correct conclusion.

Answer

The ordinary improper integral diverges to $+\infty$; Newton-Leibniz cannot be applied across $x = 1$.

Knowledge points

- interior singularities
- independent convergence on both sides
- Cauchy principal value versus ordinary convergence

Reading and method choice

The integrand is undefined at the interior point 1 and is nonnegative; the negative result is already a warning, and the interval must be split into two one-sided limits.

Detailed derivation

1. The denominator vanishes at $x = 1$, so this is an improper integral with an interior singularity.
2. The left part is $\int_0^1 \frac{dx}{(x-1)^2} = \lim_{c\to1^-} \left[-\frac{1}{x-1}\right]_0^c$.
3. Here $-\frac{1}{c-1} \to +\infty$, so the left part diverges.
4. The right part is $\int_1^2 \frac{dx}{(x-1)^2} = \lim_{d\to1^+} \left[-\frac{1}{x-1}\right]_d^2$, which also tends to $+\infty$.
5. Divergence on either side is enough; both are positive infinite, so a negative value is impossible.

Common pitfalls

- applying Newton-Leibniz across a singularity
- canceling positive and negative infinities
- reporting a symmetric truncation limit as the ordinary integral

Verification

Because the integrand is nonnegative wherever defined, every truncated integral is nonnegative; the value -2 fails the most basic sign check.

Method takeaway

An interior singularity splits the interval; divergence on either side makes the ordinary improper integral divergent.

Extension

If a symmetric principal value exists, label it explicitly as PV rather than omitting the qualifier.

Source lineage: Original synthesis / diagnosis · improper_integral_interior_singularity · reference IDs: mit-18.01sc-improper-integrals. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard | Calculation | Synthesis | Classic-method variant

Q087 · Algebraic Decay and Substitution

Problem

Evaluate $\displaystyle\int_0^{+\infty} \frac{x}{(1+x^2)^{\frac{3}{2}}}\, dx.$

Answer

1

Knowledge points

- definition on an infinite interval
- truncation limits
- convergence values and tails

Reading and method choice

With $u = 1 + x^2$, the problem becomes the standard tail integral $u^{-\frac{3}{2}}$; the exponent is below -1, so it converges at infinity.

Detailed derivation

1. First write $\lim_{R\to+\infty} \int_0^R \frac{x}{(1+x^2)^{\frac{3}{2}}} dx$.
2. Set $u = 1 + x^2$, so $du = 2x\, dx$ and the endpoints become 1 and $1 + R^2$.
3. The truncated integral is $\frac{1}{2} \int_1^{1+R^2} u^{-\frac{3}{2}} du$.
4. Its value is $1 - \frac{1}{\sqrt{1+R^2}}$.
5. As $R \to +\infty$, the second term tends to 0, so the answer is 1.

Common pitfalls

- substituting infinity as if it were an endpoint
- failing to split a two-sided infinite interval into independent limits
- confusing a symmetric principal value with ordinary convergence

Verification

An antiderivative is $-\frac{1}{\sqrt{1+x^2}}$; its endpoint difference from 0 to infinity is $0-(-1)=1$.

Method takeaway

An infinite endpoint must be truncated before taking a limit, and a two-sided integral requires both one-sided pieces to converge independently.

Extension

Vary the truncation scheme to expose the difference between a principal value and an ordinary improper integral.

Source lineage: Classic-method variant · improper_integral_infinite_interval · reference IDs: mit-18.01sc-improper-integrals. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard Proof Synthesis Classic-method variant

Q088 · Reciprocal Substitution between Infinity and Zero

Problem

Let f be continuous on $[1, +\infty)$. Prove that whenever either side converges, $\int_1^{+\infty} f(x)\,dx = \int_0^1 \frac{f\left(\frac{1}{t}\right)}{t^2}\,dt$, and show that the two sides converge or diverge together.

Answer

See the proof.

Knowledge points

- definition on an infinite interval
- truncation limits
- convergence values and tails

Reading and method choice

First substitute $x = \frac{1}{t}$ on the finite interval $[1, R]$; the infinite endpoint becomes the new singular endpoint $t = 0^+$, and the two truncated limits are identical.

Detailed derivation

1. For any $R > 1$, set $x = \frac{1}{t}$, so $dx = -\frac{1}{t^2}dt$.
2. When $x = 1$, $t = 1$; when $x = R$, $t = \frac{1}{R}$.
3. Hence $\int_1^R f(x)\,dx = \int_{\frac{1}{R}}^1 \frac{f\left(\frac{1}{t}\right)}{t^2}\,dt$.
4. Letting $R \to +\infty$ is equivalent to setting $\varepsilon = \frac{1}{R} \to 0^+$.
5. The truncated expressions are identical, so one has a finite limit exactly when the other does.
6. Taking that limit in the convergent case proves the identity.

Common pitfalls

- substituting infinity as if it were an endpoint
- failing to split a two-sided infinite interval into independent limits
- confusing a symmetric principal value with ordinary convergence

Verification

For $f(x) = x^{-p}$, the right side becomes $\int_0^1 t^{p-2}dt$, which requires $p - 2 > -1$, namely $p > 1$, matching the infinite-interval power test.

Method takeaway

An infinite endpoint must be truncated before taking a limit, and a two-sided integral requires both one-sided pieces to converge independently.

Extension

Vary the truncation scheme to expose the difference between a principal value and an ordinary improper integral.

Source lineage: Classic-method variant · improper_integral_infinite_interval · reference IDs: mit-18.01sc-improper-integrals. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard | Calculation | Synthesis | Open-text method adaptation

Q089 · Complete Parameter Classification for an Endpoint Power Integral

Problem

For real p, determine convergence and evaluate $I(p) = \int_0^1 \frac{dx}{x^p}$.

Answer

It converges exactly for $p < 1$, with $I(p) = \frac{1}{1-p}$; it diverges for $p \geq 1$.

Knowledge points

- comparison tests for improper integrals
- limit comparison
- absolute and conditional convergence

Reading and method choice

The power near 0 is $-p$; treat the logarithmic boundary case $p = 1$ separately from the power antiderivative for $p \neq 1$.

Detailed derivation

1. By definition, examine $\lim_{\varepsilon \to 0^+} \int_\varepsilon^1 x^{-p}\,dx$.
2. For $p \neq 1$, the truncated integral is $\frac{1-\varepsilon^{1-p}}{1-p}$.
3. If $p < 1$, then $1 - p > 0$ and $\varepsilon^{1-p} \to 0$, giving $\frac{1}{1-p}$.
4. If $p > 1$, then $1 - p < 0$ and $\varepsilon^{1-p} \to +\infty$, so the integral diverges to $+\infty$.
5. For $p = 1$, the truncated integral is $-\ln \varepsilon \to +\infty$.
6. Thus the convergence condition is exactly $p < 1$.

Common pitfalls

- reversing the comparison inequality
- using the finite-positive limit conclusion when the ratio tends to zero or infinity
- proving convergence without checking absolute convergence

Verification

For $p = \frac{1}{2}$ the value is 2, agreeing with Q073; $p = 1$ gives logarithmic divergence, checking the threshold.

Method takeaway

A convergence comparison must state nonnegativity, the tail or neighborhood being compared, and the benchmark integral.

Extension

Use the same benchmark to build both upper and lower comparisons and clarify the direction of sufficient and necessary conditions.

Source lineage: Open-text method adaptation · improper_integral_convergence_tests · reference IDs: mit-18.01sc-improper-integrals, openstax-calculus-v2-3.7. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard Proof Synthesis Classic-method variant

Q090 · The Limit Comparison Test

Problem

Let positive functions f, g be integrable on every finite interval $[a, R]$, and suppose $\lim_{x\to+\infty} \frac{f(x)}{g(x)} = L$ with $0 < L < +\infty$. Prove that $\int_a^{+\infty} f(x)\,dx$ and $\int_a^{+\infty} g(x)\,dx$ have the same convergence behavior.

Answer

See the proof.

Knowledge points

- comparison tests for improper integrals
- limit comparison
- absolute and conditional convergence

Reading and method choice

Use the limit definition to trap the ratio between two positive constants, producing two-sided tail comparisons; a finite initial segment does not affect convergence.

Detailed derivation

1. Take $\varepsilon = \frac{L}{2}$. The limit definition gives $M \geq a$ such that $\left|\frac{f(x)}{g(x)} - L\right| < \frac{L}{2}$ for $x \geq M$.
2. Hence $\frac{L}{2} < \frac{f(x)}{g(x)} < \frac{3L}{2}$, or $\frac{L}{2}g(x) < f(x) < \frac{3L}{2}g(x)$.
3. If $\int_a^{+\infty} g$ converges, its tail converges, and the upper comparison makes $\int_M^{+\infty} f$ converge.
4. If $\int_a^{+\infty} f$ converges, then $g(x) < \frac{2}{L}f(x)$ makes $\int_M^{+\infty} g$ converge.
5. Both integrals over the finite interval $[a, M]$ are finite and do not affect tail convergence.
6. Therefore the two improper integrals converge or diverge together.

Common pitfalls

- reversing the comparison inequality
- using the finite-positive limit conclusion when the ratio tends to zero or infinity
- proving convergence without checking absolute convergence

Verification

For $f(x) = \frac{1}{x^2+1}$ and $g(x) = \frac{1}{x^2}$, the ratio tends to 1, and both integrals converge on $[1, +\infty)$.

Method takeaway

A convergence comparison must state nonnegativity, the tail or neighborhood being compared, and the benchmark integral.

Extension

Use the same benchmark to build both upper and lower comparisons and clarify the direction of sufficient and necessary conditions.

Source lineage: Classic-method variant · improper_integral_convergence_tests · reference IDs: mit-18.01sc-improper-integrals. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard | Calculation | Synthesis | Classic-method variant

Q091 · A Logarithmically Corrected Power Integral

Problem

For real p, determine convergence and evaluate $J(p) = \displaystyle\int_e^{+\infty} \frac{dx}{x(\ln x)^p}$.

Answer

It converges exactly for $p > 1$, with $J(p) = \frac{1}{p-1}$; it diverges for $p \leq 1$.

Knowledge points

- comparison tests for improper integrals
- limit comparison
- absolute and conditional convergence

Reading and method choice

The substitution $u = \ln x$ converts the logarithmically corrected integral exactly into a power integral on an infinite interval, with lower limit 1.

Detailed derivation

1. For finite $R > e$, set $u = \ln x$, so $du = \frac{dx}{x}$.
2. The endpoints become $x = e \Rightarrow u = 1$ and $x = R \Rightarrow u = \ln R$.
3. Thus $\int_e^R \frac{dx}{x(\ln x)^p} = \int_1^{\ln R} u^{-p}\, du$.
4. As $R \to +\infty$, $\ln R \to +\infty$.
5. The infinite-interval power test gives convergence exactly for $p > 1$, with value $\frac{1}{p-1}$.

Common pitfalls

- reversing the comparison inequality
- using the finite-positive limit conclusion when the ratio tends to zero or infinity
- proving convergence without checking absolute convergence

Verification

For $p = 2$, an antiderivative is $-\frac{1}{\ln x}$, whose endpoint difference from e to infinity is $0 - (-1) = 1$, agreeing with the formula.

Method takeaway

A convergence comparison must state nonnegativity, the tail or neighborhood being compared, and the benchmark integral.

Extension

Use the same benchmark to build both upper and lower comparisons and clarify the direction of sufficient and necessary conditions.

Source lineage: Classic-method variant · improper_integral_convergence_tests · reference IDs: mit-18.01sc-improper-integrals. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard Proof Synthesis Original synthesis / diagnosis

Q092 · Conditional Convergence of an Oscillatory Integral

Problem

Prove that $\displaystyle\int_1^{+\infty} \frac{\sin x}{x}\, dx$ converges but not absolutely.

Answer

It converges conditionally.

Knowledge points

- comparison tests for improper integrals
- limit comparison
- absolute and conditional convergence

Reading and method choice

Ordinary convergence follows by controlling arbitrary tails through integration by parts; the absolute integral has a harmonic-type lower bound on intervals where sine stays away from zero.

Detailed derivation

1. For $B > A \geq 1$, integration by parts gives $\int_A^B \frac{\sin x}{x} dx = -\frac{\cos x}{x}\Big|_A^B - \int_A^B \frac{\cos x}{x^2} dx$.
2. Hence the tail magnitude is at most $\frac{1}{A} + \frac{1}{B} + \int_A^B \frac{dx}{x^2} < \frac{3}{A}$.
3. This bound tends to 0 as $A \to +\infty$, so the Cauchy tail criterion proves convergence.
4. For every integer $k \geq 1$, $|\sin x| \geq \frac{1}{2}$ on $[k\pi + \frac{\pi}{6}, k\pi + \frac{5\pi}{6}]$.
5. The absolute integral on this interval is at least $\frac{1}{2} \cdot \frac{2\pi}{3} \cdot \frac{1}{(k+1)\pi} = \frac{1}{3(k+1)}$.
6. Summing these lower bounds over disjoint intervals gives a divergent harmonic tail, so $\int_1^{+\infty} \frac{|\sin x|}{x} dx$ diverges.

Common pitfalls

- reversing the comparison inequality
- using the finite-positive limit conclusion when the ratio tends to zero or infinity
- proving convergence without checking absolute convergence

Verification

The tail estimate proves convergence and gives truncation error at most $\frac{3}{A}$ beyond A; the absolute-value lower bound explicitly rules out absolute convergence.

Method takeaway

A convergence comparison must state nonnegativity, the tail or neighborhood being compared, and the benchmark integral.

Extension

Use the same benchmark to build both upper and lower comparisons and clarify the direction of sufficient and necessary conditions.

Source lineage: Original synthesis / diagnosis · improper_integral_convergence_tests · reference IDs: mit-18.01sc-improper-integrals. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard | Parameter / synthesis / application | Synthesis | Open-text method adaptation

Q093 · The Convergence Parameter of the Gamma Integral

Problem

For real s, determine convergence of $\Gamma(s) = \int_0^{+\infty} x^{s-1}e^{-x}\,dx$.

Answer

It converges exactly for $s > 0$.

Knowledge points

- improper-integral definition of the Gamma function
- parameter range for convergence
- recurrence and scaling substitution

Reading and method choice

Split at 1: near zero, behavior is controlled by the power x^{s-1}; at infinity, exponential decay dominates.

Detailed derivation

1. On $0 < x \le 1$, $e^{-1} \le e^{-x} \le 1$, so $x^{s-1}e^{-x}$ has the same local convergence as x^{s-1}.
2. The endpoint power integral $\int_0^1 x^{s-1}dx$ converges exactly when $s > 0$.
3. Thus for $s \le 0$, the Gamma integral already diverges near 0.
4. If $s > 0$, any fixed power x^{s-1} grows more slowly than $e^{\frac{x}{2}}$; there is M such that $x^{s-1} \le e^{\frac{x}{2}}$ for $x \ge M$.
5. Hence $0 \le x^{s-1}e^{-x} \le e^{-\frac{x}{2}}$, and $\int_M^{+\infty} e^{-\frac{x}{2}}\,dx$ converges.
6. A finite middle interval does not affect convergence, so convergence occurs exactly for $s > 0$.

Common pitfalls

- omitting the parameter condition near zero
- failing to justify the boundary term at infinity
- losing a scale factor or power during substitution

Verification

For $s = 1$, the integral is $\int_0^{+\infty} e^{-x} dx = 1$; at the boundary $s = 0$, it behaves like x^{-1} near zero and diverges.

Method takeaway

The Gamma function extends factorial structure through a parameter integral, but each identity is valid only in its convergence range.

Extension

Extract scale parameters first, then use the recurrence to reduce to $\Gamma(1)$ or $\Gamma\left(\frac{1}{2}\right)$.

Source lineage: Open-text method adaptation · gamma_function_enrichment · reference IDs: mit-18.100a-gamma, openstax-calculus-v2-3.7. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard Proof Synthesis Classic-method variant

Q094 · The Gamma Recurrence

Problem

Let $s > 0$. Prove $\Gamma(s+1) = s\Gamma(s)$ and justify both boundary limits used in integration by parts.

Answer

$\Gamma(s+1) = s\Gamma(s)$.

Knowledge points

- improper-integral definition of the Gamma function
- parameter range for convergence
- recurrence and scaling substitution

Reading and method choice

Apply integration by parts to a truncated integral with $u = x^s$ and $dv = e^{-x}dx$; the key is proving $x^s e^{-x}$ tends to zero at both endpoints.

Detailed derivation

1. For $0 < \varepsilon < R$, $\int_\varepsilon^R x^s e^{-x} dx = -x^s e^{-x}\big|_\varepsilon^R + s\int_\varepsilon^R x^{s-1} e^{-x} dx$.
2. Because $s > 0$, $\varepsilon^s e^{-\varepsilon} \to 0$ as $\varepsilon \to 0^+$.
3. As $R \to +\infty$, exponential decay dominates every fixed power, so $R^s e^{-R} \to 0$.
4. By Q093, $\int_0^{+\infty} x^{s-1} e^{-x} dx$ converges for $s > 0$.
5. Letting $\varepsilon \to 0^+$ and $R \to +\infty$ eliminates the boundary term and gives $\Gamma(s+1) = s\Gamma(s)$.

Common pitfalls

- omitting the parameter condition near zero
- failing to justify the boundary term at infinity
- losing a scale factor or power during substitution

Verification

For $s = 1$, the recurrence gives $\Gamma(2) = \Gamma(1) = 1$; direct evaluation of $\int_0^{+\infty} xe^{-x} dx$ also gives 1.

Method takeaway

The Gamma function extends factorial structure through a parameter integral, but each identity is valid only in its convergence range.

Extension

Extract scale parameters first, then use the recurrence to reduce to $\Gamma(1)$ or $\Gamma\left(\frac{1}{2}\right)$.

Source lineage: Classic-method variant · gamma_function_enrichment · reference IDs: mit-18.100a-gamma. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard | Calculation | Synthesis | Classic-method variant

Q095 · Gamma Function and Factorials

Problem

Let n be a nonnegative integer. Use the definition and recurrence to find $\Gamma(n+1)$.

Answer

$\Gamma(n+1) = n!$

Knowledge points

- improper-integral definition of the Gamma function
- parameter range for convergence
- recurrence and scaling substitution

Reading and method choice

First compute $\Gamma(1)$, then apply the recurrence successively for $s = 1, 2, \dots, n$; the empty product covers $n = 0$.

Detailed derivation

1. $\Gamma(1) = \int_0^{+\infty} e^{-x} dx = 1$.
2. The recurrence gives $\Gamma(2) = 1\Gamma(1)$, $\Gamma(3) = 2\Gamma(2)$, and so on.
3. Repeated expansion yields $\Gamma(n+1) = n(n-1) \cdots 2 \cdot 1\, \Gamma(1)$.
4. Since $\Gamma(1) = 1$, $\Gamma(n+1) = n!$.
5. For $n = 0$, the statement is $\Gamma(1) = 0! = 1$, so it also holds.

Common pitfalls

- omitting the parameter condition near zero
- failing to justify the boundary term at infinity
- losing a scale factor or power during substitution

Verification

For $n = 3$, $\Gamma(4) = 3! = 6$; repeated integration by parts directly verifies $\int_0^{+\infty} x^3 e^{-x} dx = 6$.

Method takeaway

The Gamma function extends factorial structure through a parameter integral, but each identity is valid only in its convergence range.

Extension

Extract scale parameters first, then use the recurrence to reduce to $\Gamma(1)$ or $\Gamma\left(\frac{1}{2}\right)$.

Source lineage: Classic-method variant · gamma_function_enrichment · reference IDs: mit-18.100a-gamma. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard | Parameter / synthesis / application | Synthesis | Classic-method variant

Q096 · Scaling Substitution and a Gamma Value

Problem

Evaluate $\int_0^{+\infty} x^3 e^{-2x}\,dx.$

Answer

$\frac{3}{8}$

Knowledge points

- improper-integral definition of the Gamma function
- parameter range for convergence
- recurrence and scaling substitution

Reading and method choice

First use $u = 2x$ to extract all scale factors, then recognize $\Gamma(4) = 3!$; x^3 and dx together contribute 2^{-4}.

Detailed derivation

1. Set $u = 2x$, so $x = \frac{u}{2}$ and $dx = \frac{1}{2}du$.
2. The endpoints remain $0, +\infty$.
3. The integral becomes $\frac{1}{2^4}\int_0^{+\infty} u^3 e^{-u}du$.
4. The remaining integral is $\Gamma(4) = 3! = 6$.
5. Therefore the value is $\frac{6}{16} = \frac{3}{8}$.

Common pitfalls

- omitting the parameter condition near zero
- failing to justify the boundary term at infinity
- losing a scale factor or power during substitution

Verification

Repeated integration by parts gives $I_3 = \frac{3}{2}I_2 = \frac{3}{2}I_1 = \frac{3}{4}I_0$ with $I_0 = \frac{1}{2}$, again yielding $I_3 = \frac{3}{8}$.

Method takeaway

The Gamma function extends factorial structure through a parameter integral, but each identity is valid only in its convergence range.

Extension

Extract scale parameters first, then use the recurrence to reduce to $\Gamma(1)$ or $\Gamma\left(\frac{1}{2}\right)$.

Source lineage: Classic-method variant · gamma_function_enrichment · reference IDs: mit-18.100a-gamma. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Challenge Calculation Challenge Classic-method variant

Q097 · A Half-integer Gamma Value

Problem

Given $\Gamma\left(\frac{1}{2}\right)=\sqrt{\pi}$, evaluate $\Gamma\left(\frac{5}{2}\right)$.

Answer

$\frac{3\sqrt{\pi}}{4}$

Knowledge points

- improper-integral definition of the Gamma function
- parameter range for convergence
- recurrence and scaling substitution

Reading and method choice

Apply the recurrence twice, reducing the argument from $\frac{5}{2}$ to $\frac{3}{2}$ and then $\frac{1}{2}$, multiplying by the preceding argument each time.

Detailed derivation

1. The recurrence gives $\Gamma\left(\frac{5}{2}\right)=\frac{3}{2}\Gamma\left(\frac{3}{2}\right)$.
2. Applying it again, $\Gamma\left(\frac{3}{2}\right)=\frac{1}{2}\Gamma\left(\frac{1}{2}\right)$.
3. Insert the given value $\Gamma\left(\frac{1}{2}\right)=\sqrt{\pi}$.
4. Thus $\Gamma\left(\frac{5}{2}\right)=\frac{3}{2}\cdot\frac{1}{2}\sqrt{\pi}=\frac{3\sqrt{\pi}}{4}$.

Common pitfalls

- omitting the parameter condition near zero
- failing to justify the boundary term at infinity
- losing a scale factor or power during substitution

Verification

By definition, $\Gamma\left(\frac{5}{2}\right)=\int_0^{+\infty}x^{\frac{3}{2}}e^{-x}dx>0$; the closed form is positive and approximately 1.329.

Method takeaway

The Gamma function extends factorial structure through a parameter integral, but each identity is valid only in its convergence range.

Extension

Extract scale parameters first, then use the recurrence to reduce to $\Gamma(1)$ or $\Gamma\left(\frac{1}{2}\right)$.

Source lineage: Classic-method variant · gamma_function_enrichment · reference IDs: mit-18.100a-gamma. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Challenge Calculation Challenge Classic-method variant

Q098 · A Factorial Integral with Exponential Scaling

Problem

Let $a > 0$ and let m be a nonnegative integer. Evaluate $I_m(a) = \int_0^{+\infty} x^m e^{-ax}\,dx$.

Answer

$I_m(a) = \frac{m!}{a^{m+1}}$

Knowledge points

- improper-integral definition of the Gamma function
- parameter range for convergence
- recurrence and scaling substitution

Reading and method choice

$a > 0$ ensures exponential decay; after $u = ax$, the power and differential together contribute $a^{-(m+1)}$.

Detailed derivation

1. Set $u = ax$, so $x = \frac{u}{a}$ and $dx = \frac{1}{a}du$.
2. Because $a > 0$, the endpoints $0, +\infty$ remain $0, +\infty$.
3. Thus $I_m(a) = \frac{1}{a^{m+1}} \int_0^{+\infty} u^m e^{-u} du$.
4. The remaining integral is $\Gamma(m+1) = m!$.
5. Therefore $I_m(a) = \frac{m!}{a^{m+1}}$.

Common pitfalls

- omitting the parameter condition near zero
- failing to justify the boundary term at infinity
- losing a scale factor or power during substitution

Verification

For $m = 0$, the formula gives $\int_0^{+\infty} e^{-ax} dx = \frac{1}{a}$, agreeing with Q077 for $a > 0$.

Method takeaway

The Gamma function extends factorial structure through a parameter integral, but each identity is valid only in its convergence range.

Extension

Extract scale parameters first, then use the recurrence to reduce to $\Gamma(1)$ or $\Gamma\left(\frac{1}{2}\right)$.

Source lineage: Classic-method variant · gamma_function_enrichment · reference IDs: mit-18.100a-gamma. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Challenge Proof Challenge Classic-method variant

Q099 · Two-endpoint Convergence of a Beta-type Integral

Problem

For real α, β, prove that $\int_0^{+\infty} \frac{x^{\alpha-1}}{(1+x)^{\alpha+\beta}}\,dx$ converges if and only if $\alpha > 0$ and $\beta > 0$.

Answer

It converges exactly when $\alpha > 0$ and $\beta > 0$.

Knowledge points

- comparison tests for improper integrals
- limit comparison
- absolute and conditional convergence

Reading and method choice

Split at 1; use limit comparison with $x^{\alpha-1}$ near zero and with $x^{-\beta-1}$ at infinity, and require both endpoint conditions.

Detailed derivation

1. Let $h(x) = \frac{x^{\alpha-1}}{(1+x)^{\alpha+\beta}} > 0$.
2. As $x \to 0^+$, $\frac{h(x)}{x^{\alpha-1}} = (1+x)^{-(\alpha+\beta)} \to 1$.
3. Thus local convergence matches $\int_0^1 x^{\alpha-1}dx$, requiring $\alpha > 0$.
4. As $x \to +\infty$, $\frac{h(x)}{x^{-\beta-1}} = \left(\frac{x}{1+x}\right)^{\alpha+\beta} \to 1$.
5. Thus tail convergence matches $\int_1^{+\infty} x^{-\beta-1}dx$, requiring $\beta > 0$.
6. Both pieces must converge, so the necessary and sufficient conditions are $\alpha > 0$ and $\beta > 0$.

Common pitfalls

- reversing the comparison inequality
- using the finite-positive limit conclusion when the ratio tends to zero or infinity
- proving convergence without checking absolute convergence

Verification

For $\alpha = \beta = 1$, the integral becomes $\int_0^{+\infty} \frac{dx}{(1+x)^2} = 1$; setting either parameter to 0 creates the critical x^{-1} behavior.

Method takeaway

A convergence comparison must state nonnegativity, the tail or neighborhood being compared, and the benchmark integral.

Extension

Use the same benchmark to build both upper and lower comparisons and clarify the direction of sufficient and necessary conditions.

Source lineage: Classic-method variant · improper_integral_convergence_tests · reference IDs: mit-18.01sc-improper-integrals. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Challenge | Parameter / synthesis / application | Challenge | Original synthesis / diagnosis

Q100 · A Two-parameter Gamma Scaling Integral

Problem

For real parameters a, b, completely classify and evaluate $I(a,b) = \int_0^{+\infty} x^{a-1}e^{-bx}\,dx$.

Answer

It converges exactly when $a > 0$ and $b > 0$; then $I(a,b) = \frac{\Gamma(a)}{b^a}$.

Knowledge points

- improper-integral definition of the Gamma function
- parameter range for convergence
- recurrence and scaling substitution

Reading and method choice

Parameter a controls the power behavior near zero, while b controls the exponential tail; only when both independent conditions hold does scaling produce a Gamma value.

Detailed derivation

1. Near 0, $e^{-bx} \to 1$, so the integrand is comparable to x^{a-1}; this requires exactly $a > 0$.
2. If $b > 0$, the exponential decays at infinity and dominates every fixed power, so the tail converges.
3. If $b = 0$, the tail is a power integral requiring $a < 0$, which contradicts the near-zero condition $a > 0$; no value of a works.
4. If $b < 0$, write $b = -c$ with $c > 0$; then $e^{-bx} = e^{cx}$ grows exponentially, so the tail diverges for every real a.
5. Thus the necessary and sufficient conditions are $a > 0$ and $b > 0$.
6. In that range, set $u = bx$, so $dx = \frac{1}{b}du$ and $x^{a-1} = b^{1-a}u^{a-1}$.
7. Therefore $I(a,b) = b^{-a}\int_0^{+\infty} u^{a-1}e^{-u}du = \frac{\Gamma(a)}{b^a}$.

Common pitfalls

- omitting the parameter condition near zero
- failing to justify the boundary term at infinity

- losing a scale factor or power during substitution

Verification

For $a = 1, I(1, b) = \frac{1}{b}$, matching the exponential integral; for $a = m + 1$, it gives $\frac{m!}{b^{m+1}}$, agreeing with Q098.

Method takeaway

The Gamma function extends factorial structure through a parameter integral, but each identity is valid only in its convergence range.

Extension

Extract scale parameters first, then use the recurrence to reduce to $\Gamma(1)$ or $\Gamma\left(\frac{1}{2}\right)$.

Source lineage: Original synthesis / diagnosis · gamma_function_enrichment · reference IDs: mit-18.100a-gamma. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.